

Break It or Take It?

Best Methods for Removing Indirect Restorations

Madeline Crumpton, Navid Khorashahi, Andrew Mancao, Tariq Mohammed,
Austin Nguyen, Ryan O'Donnell, Anthony Turner II, Ethan Walden

Faculty Mentor: Dr. Britton and Dr. Harris



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Why Does This Matter?

Ryan O'Donnell



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Importance of Material-Based Crown Removal

- Choose the safest and most efficient removal technique
- Minimize the risk of pulpal injury or unnecessary tooth structure loss
- Reduce chair time and improve patient comfort
- Anticipate how each material will respond during sectioning or debonding



You're a *patient*. You have a failing zirconia crown. The dentist says they have to cut your crown off and remake it.

Question: What matters most to you — tooth preservation, speed, or comfort?





M.C. "Preserving existing tooth structure"

W.W. "Time doesn't matter. Discomfort or pain does. I would think most people would feel the same."



Behavioral Objectives



Behavioral Objective 1: Describe the biological and mechanical risks of removing indirect restorations, including potential damage to enamel, dentin, pulp, and supporting tissues.

Behavioral Objective 2: Differentiate between restorations that are bonded adhesively versus those luted conventionally, and explain how the mode of retention influences removal strategies.

Behavioral Objective 3: Explain how restorative material properties (such as brittleness, hardness, translucency, and fracture behavior) affect the selection and effectiveness of removal techniques.



Behavioral Objectives



Behavioral Objective 4: Compare and contrast varied techniques for removing crowns, onlays, and other indirect restorations (e.g., sectioning with burs, ultrasonic vibration, crown pullers, laser-assisted methods) by explaining their mechanisms of action and contrasting their advantages and limitations based on scientific evidence.

Behavioral Objective 5: Apply a biologically informed clinical decision tree to select the most appropriate removal method, integrating material type, retention strategy, and mechanical/optical properties while prioritizing tooth preservation and patient comfort.

BO1: Describe the biological and mechanical risks of removing indirect restorations, including potential damage to enamel, dentin, pulp, and supporting tissues.

Madeline Crumpton

Faculty Mentors: Dr. Britton and Dr. Harris



Search Strategy

P: Patient with an indirect restorations (crowns, onlays, inlays) that requires removal

I: Removal of indirect restorations

C: No removal of indirect restorations

O: Biological and mechanical risks/damage (to enamel, dentin, pulp, and surrounding tissues).

Databases used

Pubmed

Google Scholar

Keywords

indirect restoration removal risks, damage to tooth structure during crown removal, failed crown and bridge removal, iatrogenic trauma, mechanical and biological damage. etc.

Pubmed search string

(failed crown and bridge removal)
(removal of FPDs) AND (iatrogenic
heat) OR (pulpal trauma)
Many search strings yielded 0
results

Search results

8 results-> 2 results

Highest level of evidence

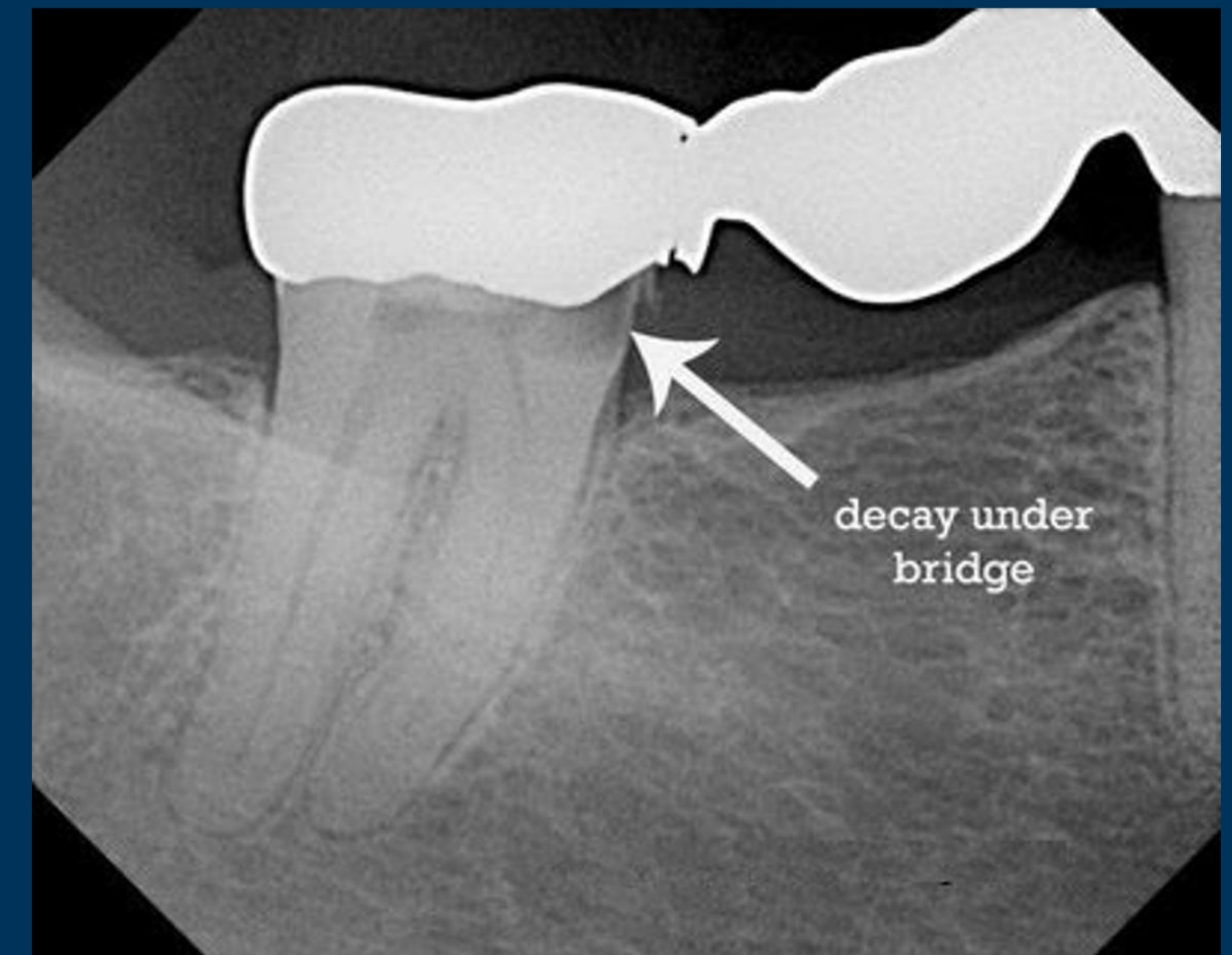
Review articles

Pre-Op Considerations

1. Comprehensive Exam & Radiographs
2. Medical Contraindications
3. Periodontal Status
4. Intra-Oral Access
5. Status of Underlying Core
6. Cement-luted vs. Bonded
7. Crown and Bridge Materials
8. Educate Patient

Primary Objectives:

- Preserve remaining tooth structure
- Maintain pulp vitality
- Protect surrounding periodontal tissues



The Challenge of Removal

The removal of permanently cemented indirect restorations (crowns, bridges, onlays, veneers) is often an “unpleasant, anxiety-ridden, risky subject for both the dentist and patient.”

Table 1. Classification of causes of crown and bridge failures.

Biological	Mechanical	Aesthetical
1.Caries	1. Cementation failure	1.Colour
2.Endodontic treatment.	2. Defective margins	2. Contour.
3.Endodontic re-treatment	3.Post and core failure under crowns/bridges.	
4.Periodontal	4. Precision attachment breakages.	
5.Occlusion	5. Fractured porcelain facings	
6. Metal allergies.		

The removal process inherently endangers the underlying tooth structure.



Risk Categorization



Mechanical Risks:

Physical damage to the hard tissues of the tooth or adjacent structures caused by instrumentation.

- ★ Enamel fractures and cracks
- ★ Dentin gouging
- ★ Core buildup fracture
- ★ Root fractures
- ★ Damage to adjacent teeth
- ★ Damage to opposing teeth

Biological Risks:

Physiological damage to the vital soft tissues of the tooth and supporting periodontium.

- ★ Thermal injury to the pulp
- ★ Irreversible pulpitis & Necrosis
- ★ Aspiration of crown and/or fractured burs
- ★ Swallowing pieces of fractured crown
- ★ Soft tissue laceration/trauma



Mechanical Risks: Enamel & Dentin



Risk	Cause	Considerations	Mitigation Strategies
Enamel Chipping and Fractures	Micro-cracks: High speed vibration and pressure can induce craze lines and fractures in the remaining cervical enamel. Chipping: Aggressive prying forces from crown removers can shear off enamel margins.	Recurrent caries often compromise underlying tooth structure/remaining enamel making it more prone to fracture especially at the margins.	Do not use excess pressure Be careful when using prying forces at cervical margins
Dentin Gouging	Difficulty distinguishing between resin cement and dentin can lead to over-preparation and gouging of the axial walls.	Excessive air-drying during removal to distinguish cement vs. dentin as well as during bonding removal can collapse collagen fibers.	Let the bur do the work and Do not use excess pressure when sectioning Take breaks, blow air and water, and re-assess with adequate light and visualization Be aware that desiccation can occur
Core Fracture	Mechanical forces from "crown tappers" or sliding hammers can fracture the composite or amalgam core buildup, necessitating complex retreatment.	Core material (amalgam, composite, CompCore, post & core) may be hard to distinguish PFM from amalgam core	Do not use excessive force when using mechanical pullers Utilize adequate light and visualization
Root/Tooth Fracture	Excessive torquing can result in horizontal root fracture, breaking off parts of the tooth	Endodontically treated teeth are more brittle and therefore more likely to fracture when using force.	No excessive force or torquing Utilize gentle, wiggling motions

Damage to Adjacent & Opposing Teeth

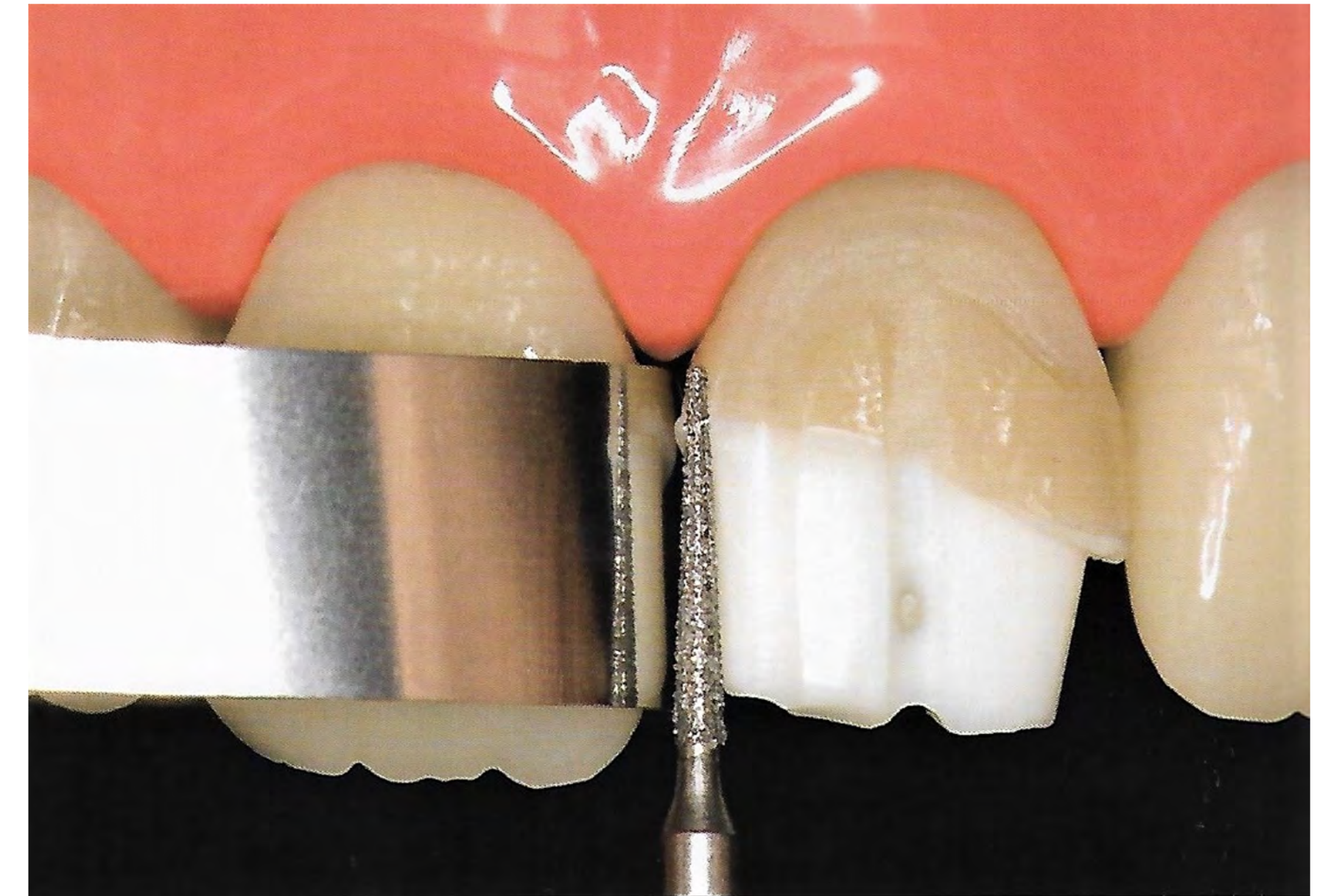


Adjacent teeth

- Research indicates 73% of adjacent teeth are nicked or damaged during restorative procedures including crown removal.
- The fracture of adjacent teeth especially those with **tight interproximal contacts** → fracture of adjacent marginal ridge/cusp or fracture of porcelain if adjacent tooth is PFM.
- **Prevention:** Check interproximal contacts with floss. Use metal matrix bands or wedges to shield adjacent teeth.

Opposing teeth

- Be aware when using prying instruments such as mechanical pullers and even crown spreaders of your patient's ability to open and their opposing teeth to avoid fracture if instruments were to dislodge suddenly. Protect opposing dentition with a finger, cotton roll or gauze.

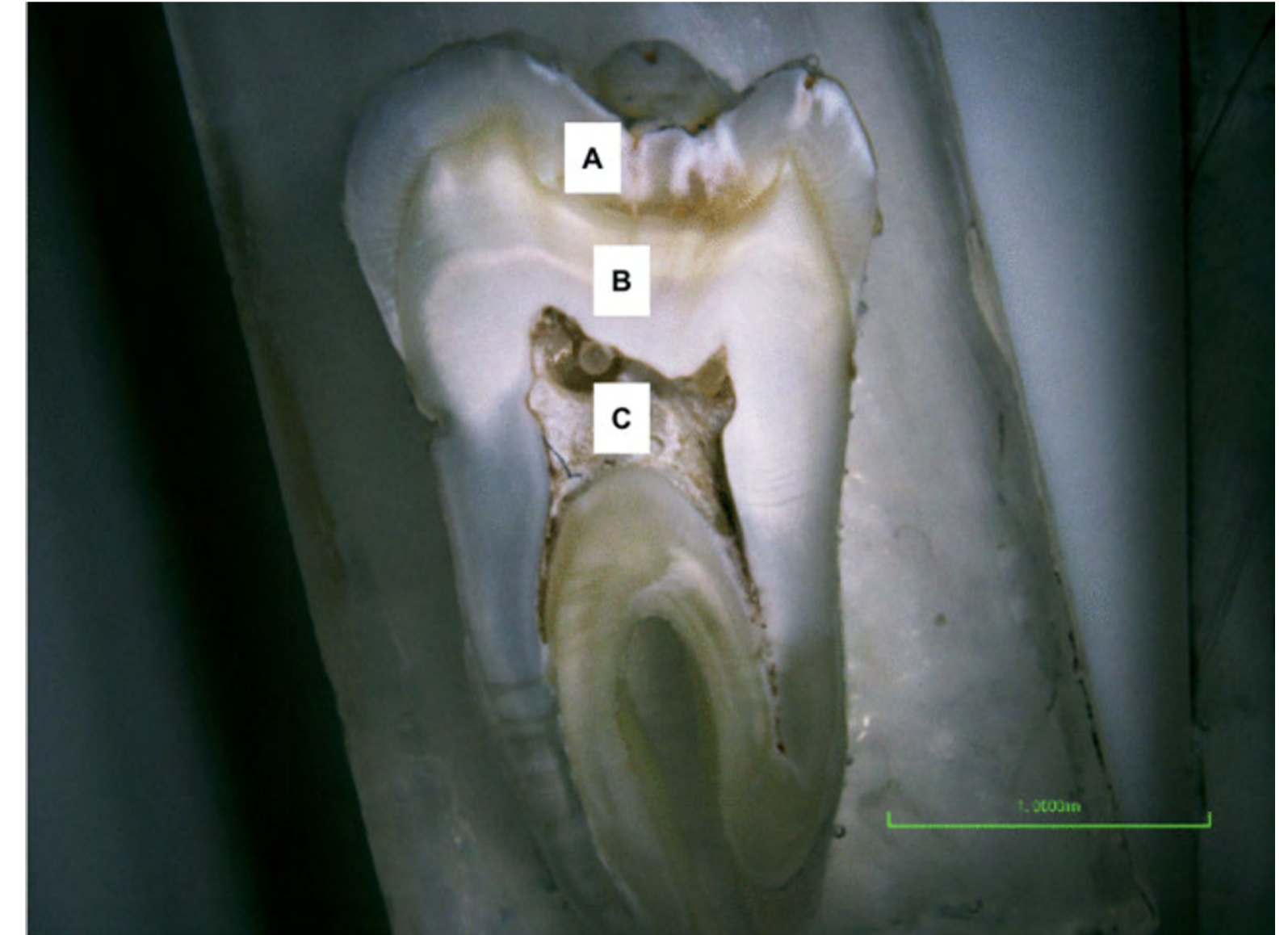


Biological Risks: The Pulp

Thermal Trauma

Heat generation is the primary biological threat to pulpal health during removal procedures as cutting through metal or zirconia crowns with diamond or carbide burs generates intense frictional heat.

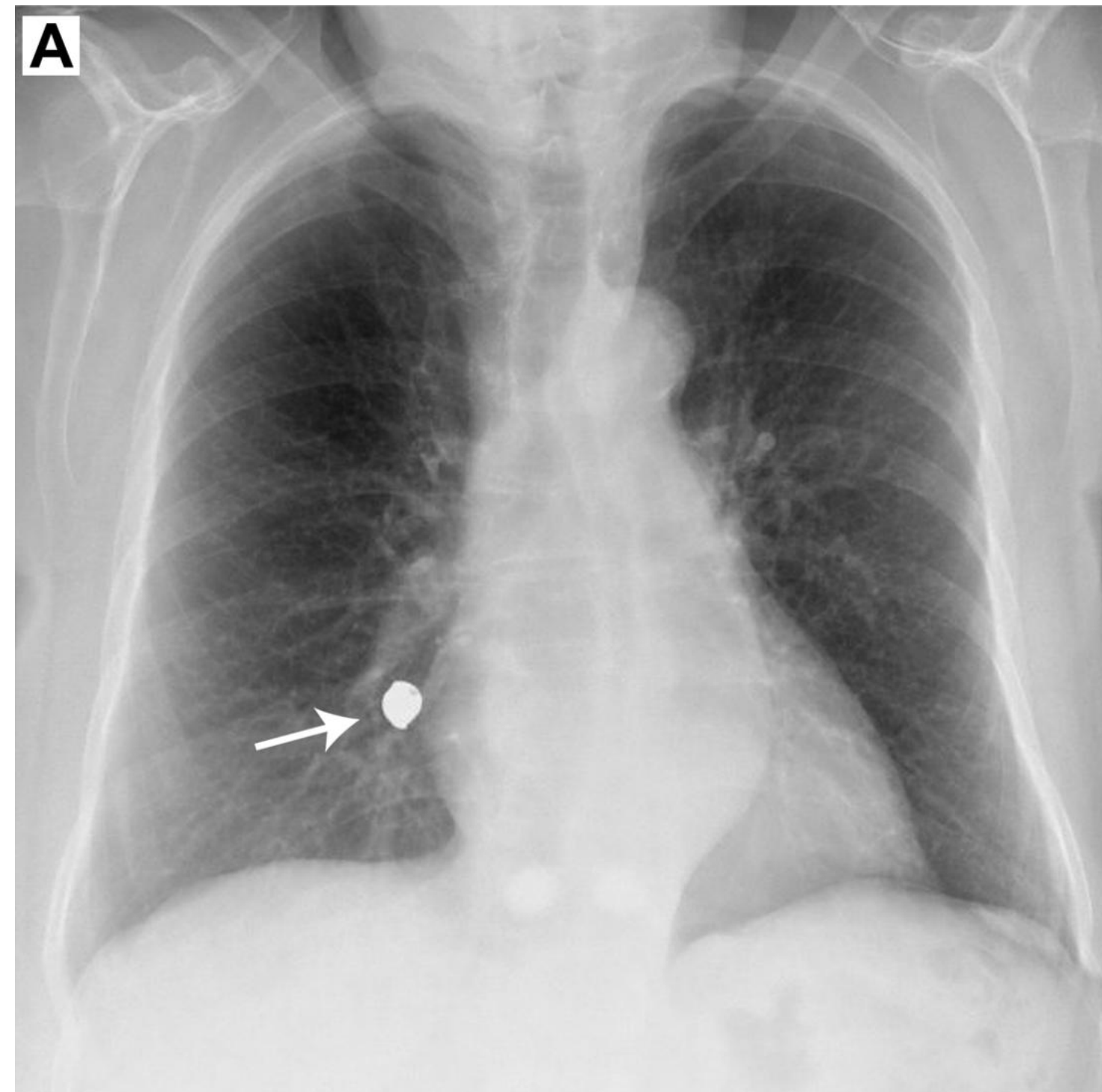
- **Critical threshold:** An intra-pulpal temperature increase of only **5.5°C** can cause irreversible pulpitis or necrosis.
- **Desiccation:** Lack of adequate water coolant accelerates thermal transfer to the odontoblasts.
- The **use of adequate water coolant** in high and slow speed handpieces did not heat the pulp past critical thresholds.
- Zirconia crown preps require more tooth be removed which means **less residual dentin thickness is left** to protect the pulp from temperature changes.



Biological Risks: Aspiration & Ingestion

Occurs when pieces of crown, tooth, or fractured burs are swallowed and go down your trachea to your airway/lungs (aspiration) or down your esophagus to your digestive tract (ingestion).

Prevented by floss ligation of pontics when sectioning bridges, throat partitions, adequate high volume suction, advising patient not to swallow, and even rubber dams.



Soft Tissue Trauma

Soft tissue lacerations of the tongue, lips, and cheeks occur most often when prepping labial and lingual surfaces of crowns when sectioning.

Damage to the tongue, lips, cheeks, and gingiva can be prevented by retraction with an aspirator tip, mouth mirror or saliva ejector.

- Advise patient to be very still and avoid sudden movements.
- Let the bur do the work and always have a firm finger rest as bur can slip or jump.
- Be aware of heat as handpieces can cause burns!



Conclusion



- The removal of indirect restorations is a routine dental procedure that can be **challenging and unpredictable** as you often do not know what's under the crown you are removing.
- Dentists must choose the appropriate removal method based on the restoration type and tooth status as well as prioritizing **THERMAL CONTROL** and **STRUCTURAL PRESERVATION** to ensure long-term success as removing the old crown is only your first step in fabricating a new crown for your patient.
 - Remember to always use **copious amounts of water and air for cooling**, HVE suction, adequate light and visualization, and always make the patient aware of what is going on as this can be a loud and uncomfortable procedure.



Every dentist has to make judgment calls about how far to go when removing a restoration.

Where's the ethical line in the sand?





Dr. Glenn Maron, GA BOD member

- Written informed consent is preferred for medico-legal protection.
- Discuss all treatment including no treatment.
- Patients make an informed decision after understanding risks, benefits, and alternatives.





Dr. Nancy Young, GA BOD member

- Well-being above procedural convenience.
- True informed consent = transparent dialogue about potential outcomes.
- Documentation and follow-up reflect a dentist's integrity, accountability, and commitment to long-term patient welfare.



BO2: Differentiate between restorations that are bonded adhesively versus those luted conventionally, and explain how the mode of retention influences removal strategies.

Tariq Mohammed

Faculty Mentors: Dr. Britton and Dr. Harris

Search Strategy



PICO:

P: Patients with indirect restorations

I: Adhesively bonded restorations

C: Conventionally luted restorations

O: Selection of removal technique

Databases used

Pubmed

Google Gemini

Keywords

Luted, Adhesively Bonded,
Crown, Veneer, Implant
Crown, Cement, Crown
Removal

Pubmed search string

((Luted) AND (Adhesively
Bonded))
AND (Crown Removal)

Search results

Filtered by: 10 years - 22
results

Highest level of evidence

Systematic review





POP QUIZ

LOG INTO D2L > DENTAL MATERIALS > EXAMS/QUIZZES

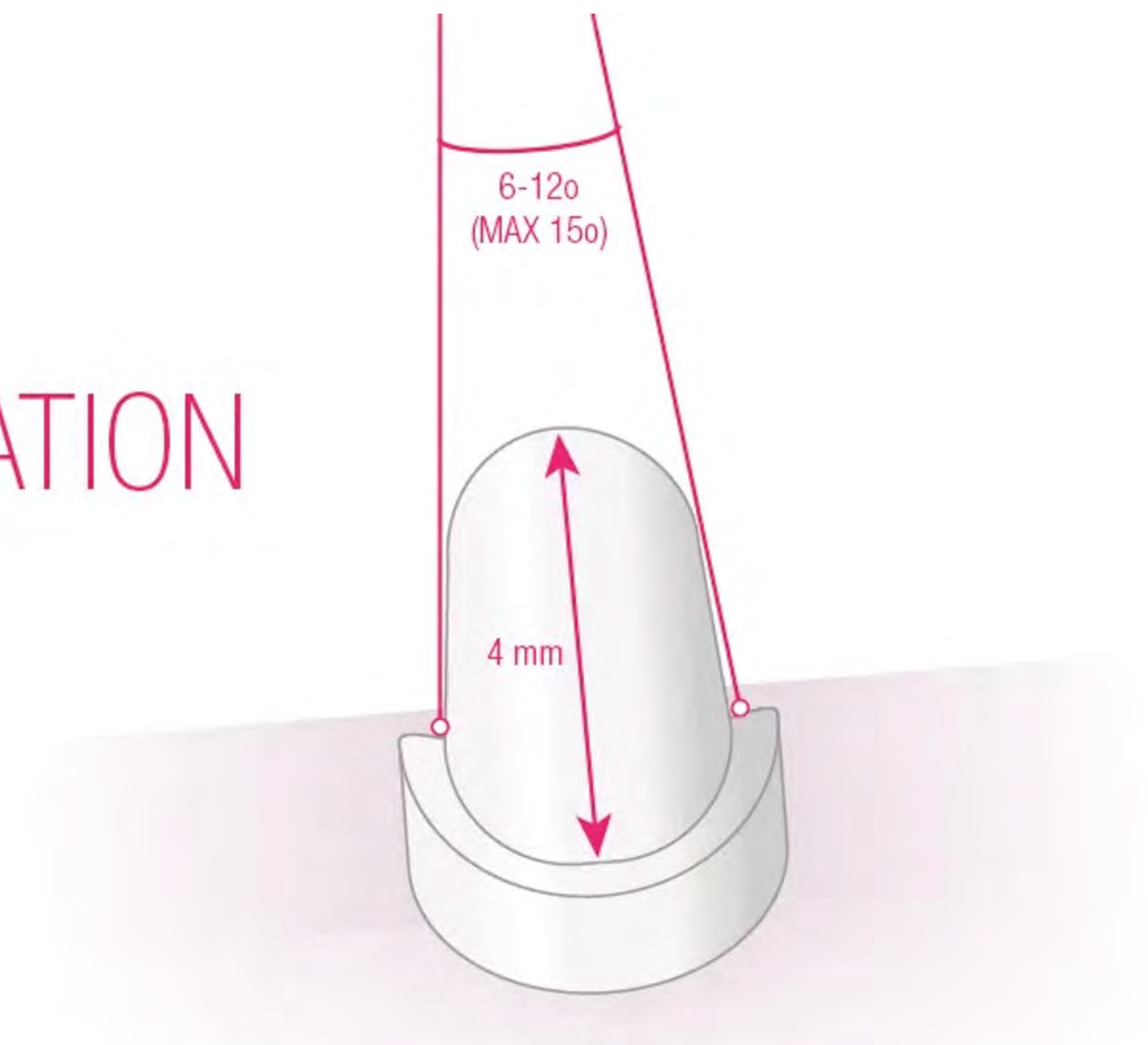
Just kidding, now that we have everyone's attention...



Conventional Luting:

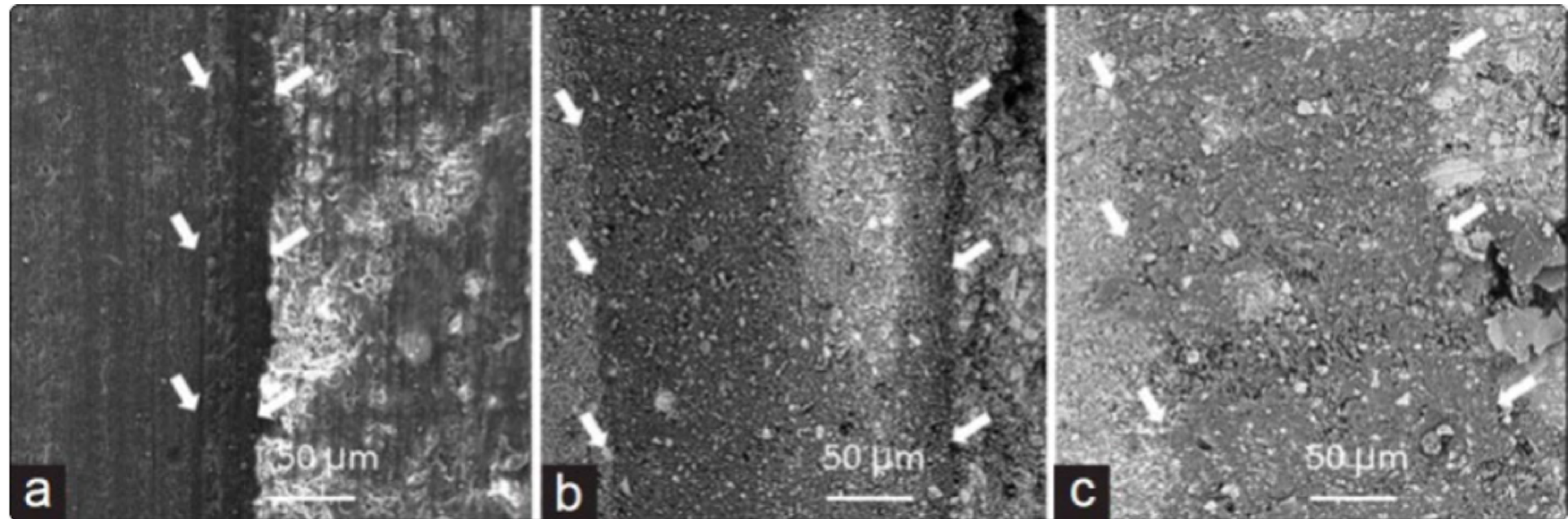
- Mechanical Retention
 - No True Chemical Bonding
- Common Cements:
 - RMGI, GIC, Zinc Phosphate
- Typically Easier Retrieval/Removal
 - Low Brittleness
 - Low Hardness
 - Low Fracture Toughness

TOOTH
PREPARATION



Google Images

SEM Microstructure of GI Cement



SEM Microstructure of GI Cement

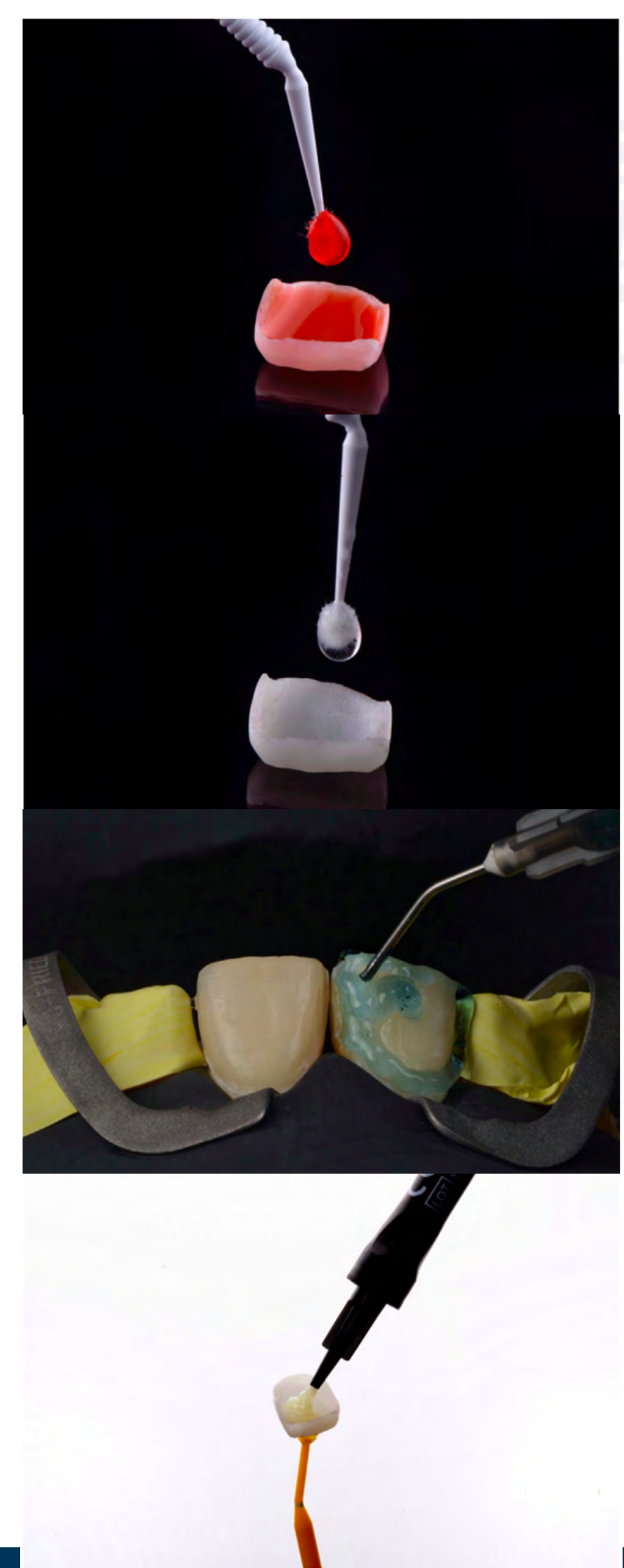


Fig. 2. Microcracks formed in the surface of a glass ionomer cement, x 1000 (Aspa[®]), indicated by arrows.



Adhesive Bonding

- Chemical and Micromechanical Retention
 - Etching/Priming to Create Hybrid Layer
 - MDP Monomers for Zirconia
- Common Cements:
 - Panavia V5 – MDP-Based Dual-Cure Resin
 - RelyX Unicem 2 – Self-Adhesive Resin Cement
- More Difficult Removal
 - Stronger Retention
 - High Hardness



Microstructure of Resin Cement

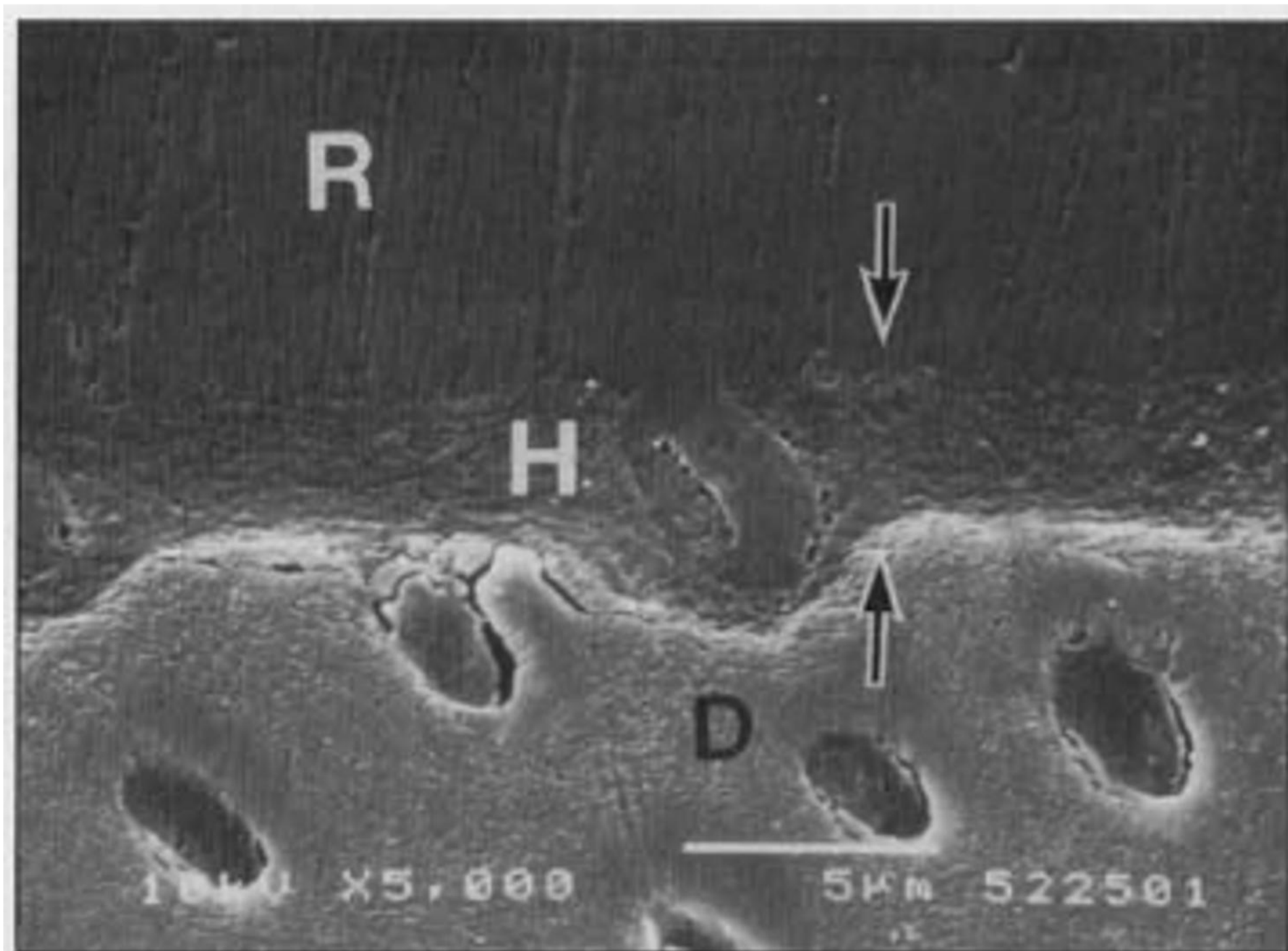


Figure 2b. Cross-sectional SEM (5000x) view of specimen depicted in Fig. 1b after resin-bonding. HA-polished dentin was pre-treated with 20P-30H for 60 s. Arrows represent the thickness of the hybrid layer. D = dentin; H = hybrid layer; R = resin.

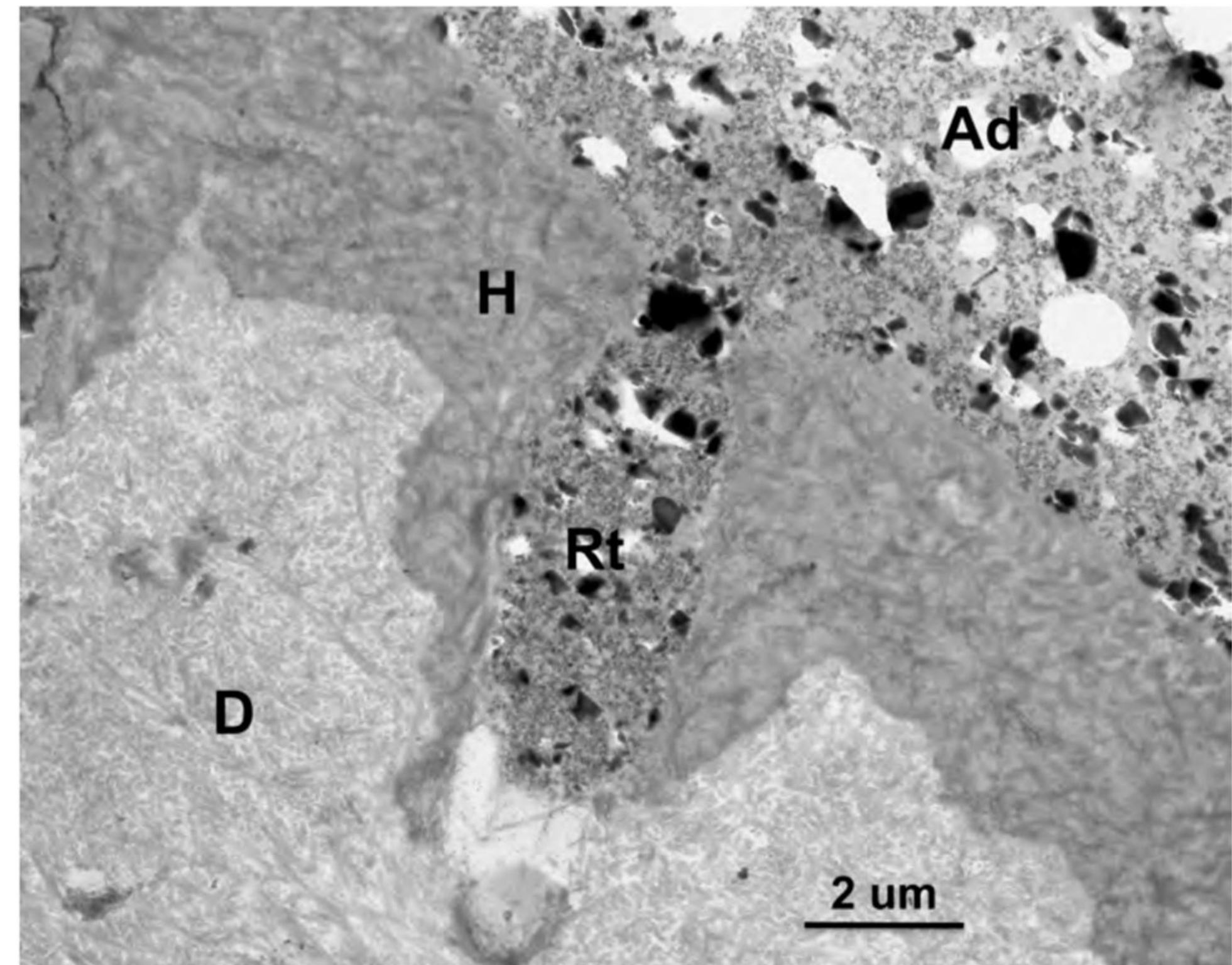


Fig. 8. TEM micrograph of the adhesive-dentin interface formed by the 3-step ER adhesive OptiBond FL (Kerr). The particle-filled hydrophobic bonding adhesive resulted in filled resin tags (Rt). Ad – adhesive; H – hybrid layer; D – dentin. Original magnification = 6,000X.



Comparing the Microstructure of RMGI vs Resin Cement



RMGI

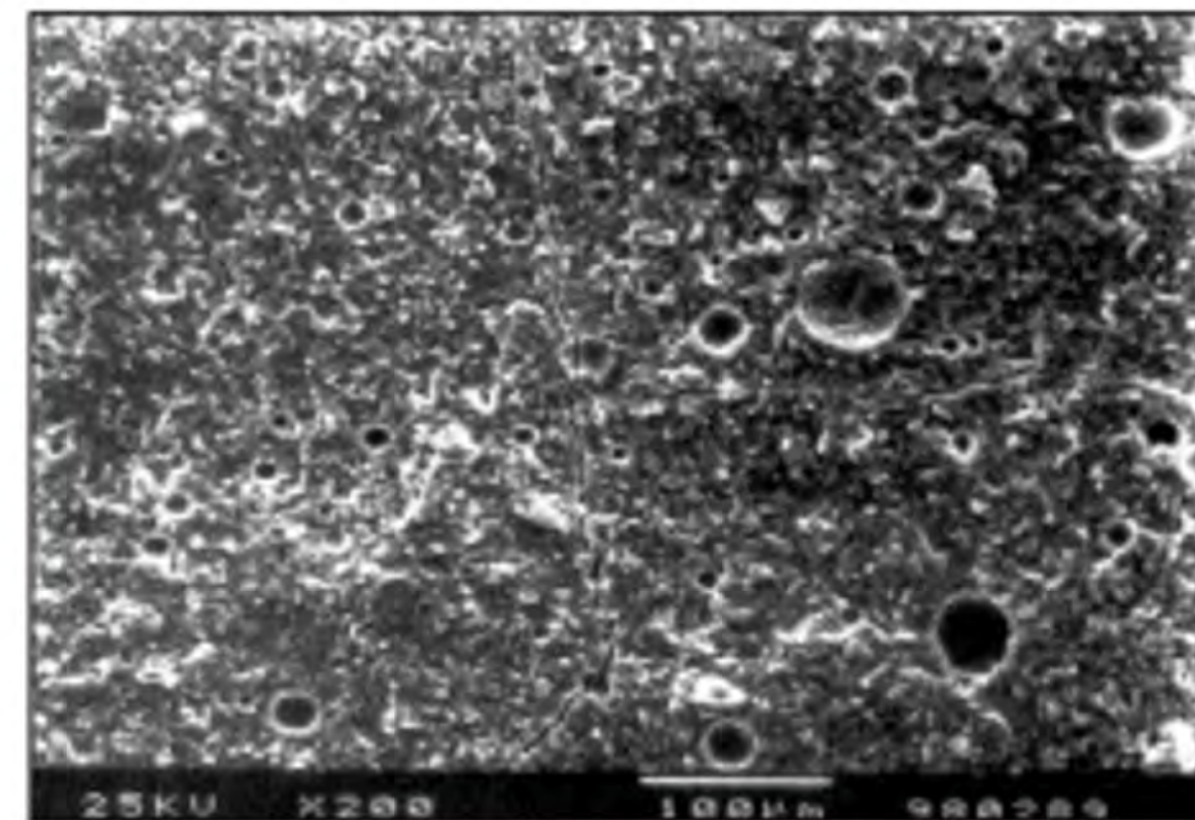
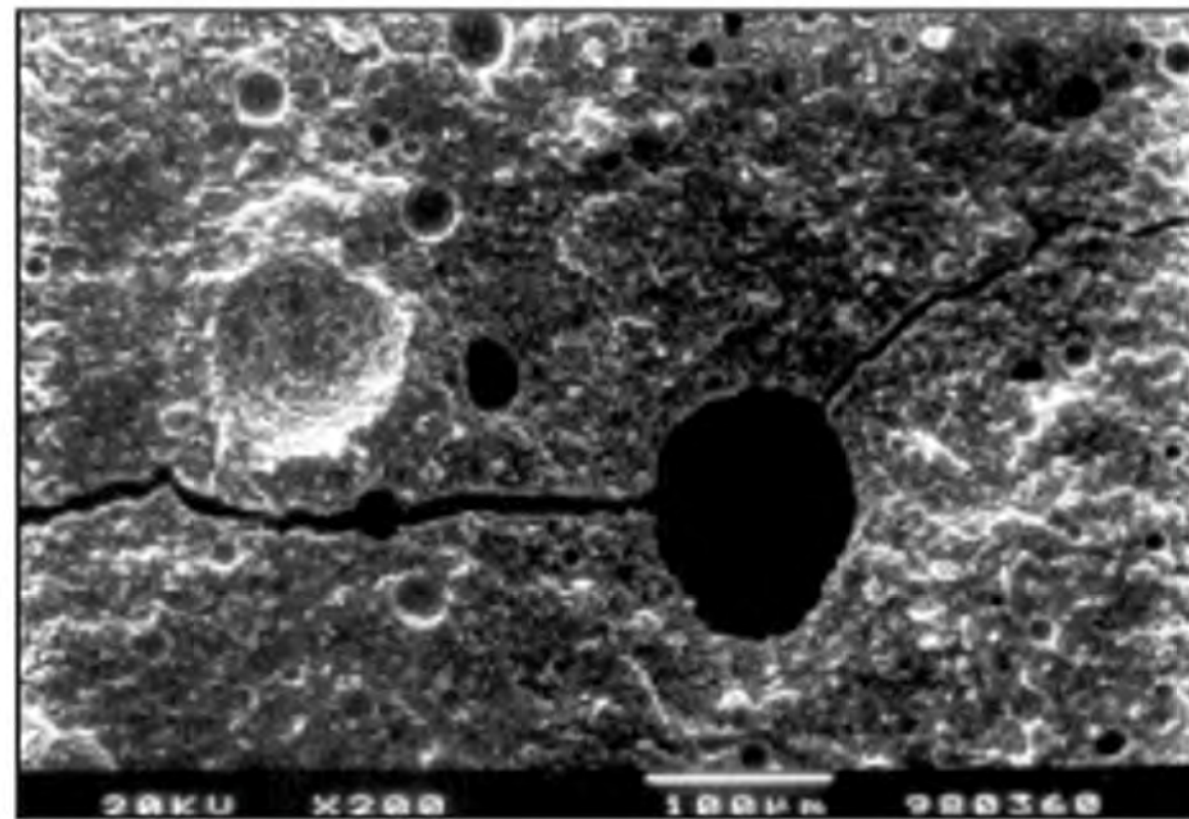


Figure 1. Representative scanning electron micrographs (SEM) taken from the fracture surface of the bending specimens of Vitremer stored in water at 37 °C for (a) 1 h, and (b) 1 week

Resin

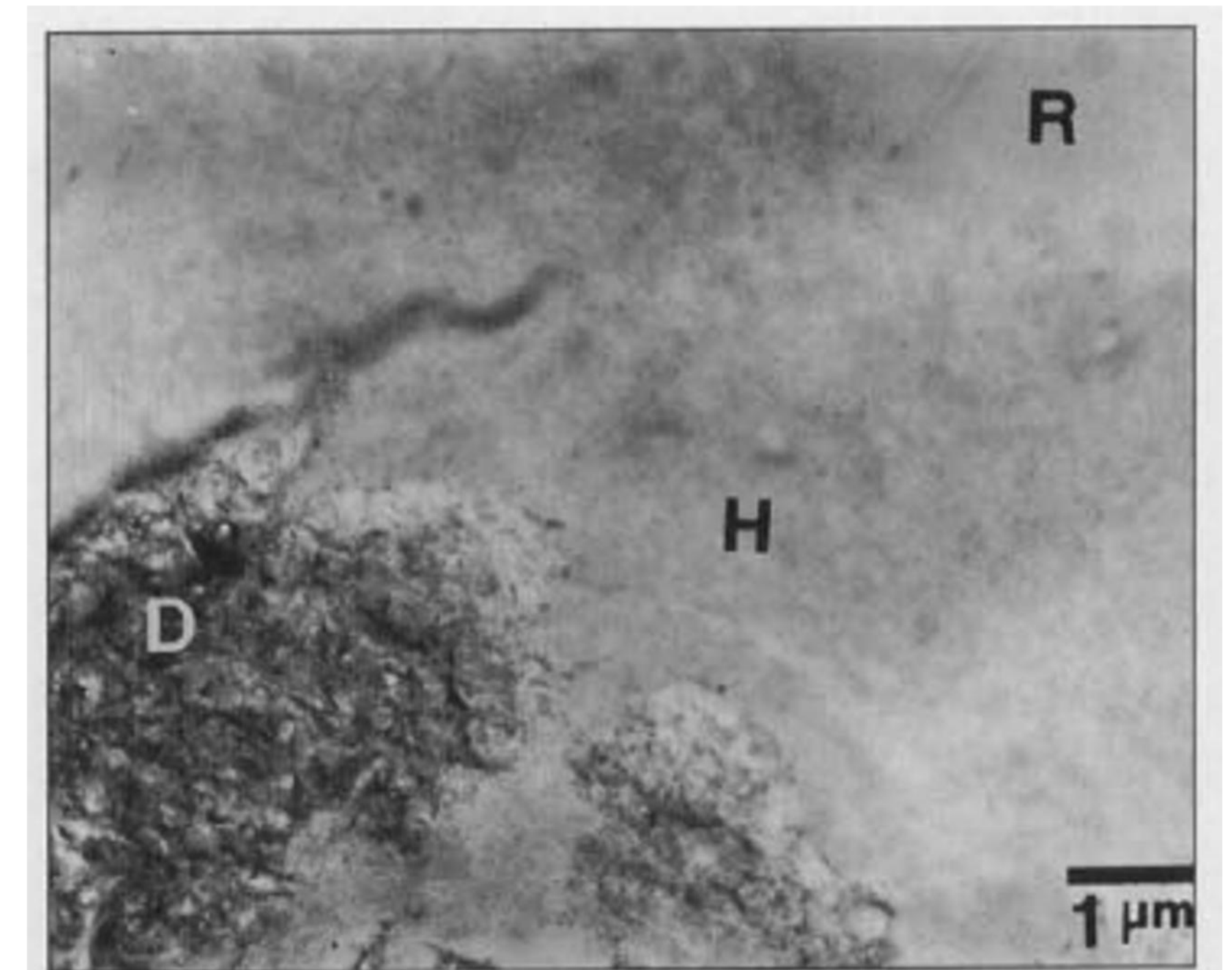
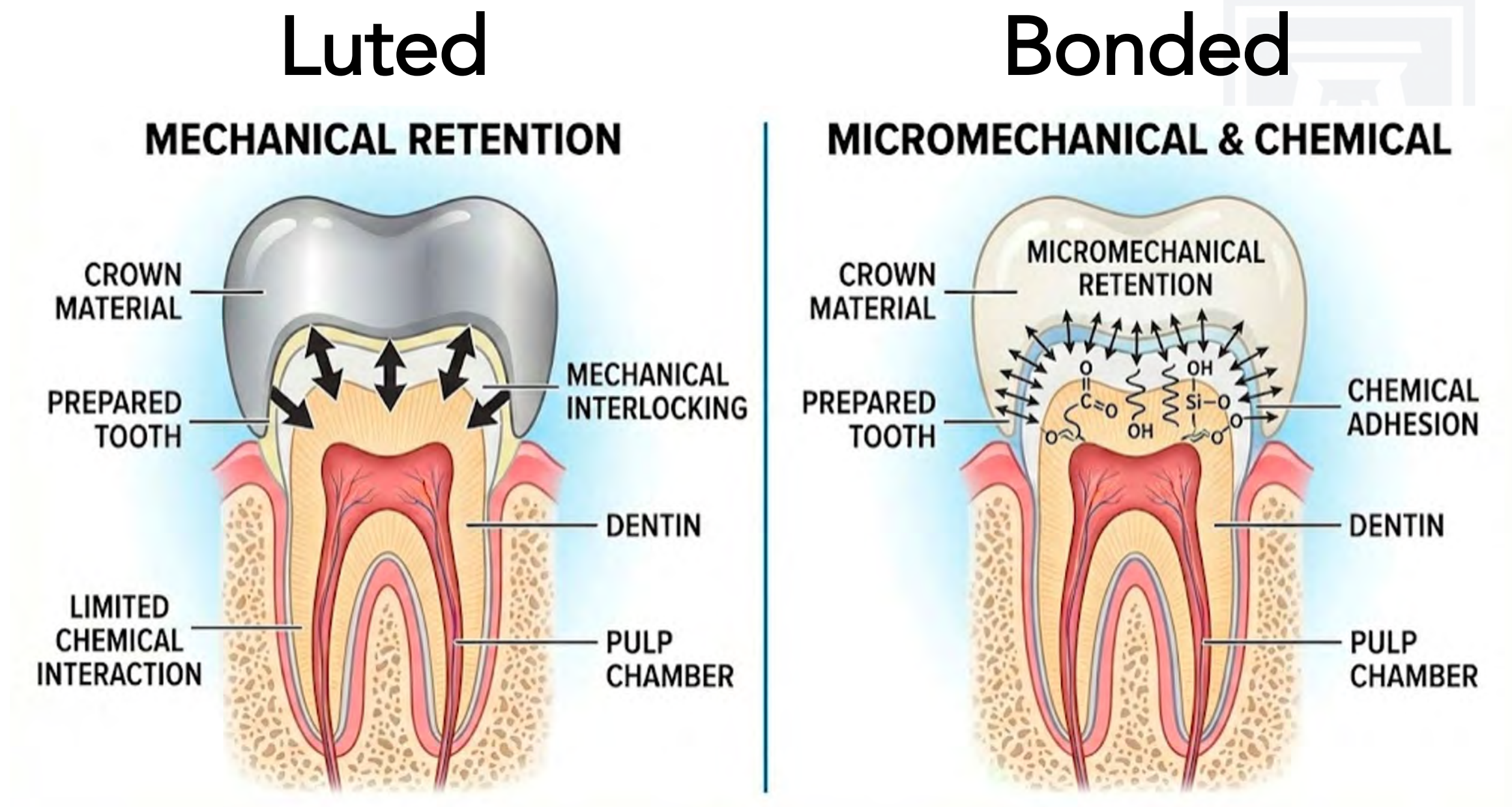


Figure 5b. Ultra-thin TEM view of HA-polished, 20P-30H pre-treated (60 s), resin-bonded specimen. D = dentin; H = hybrid layer; R = resin.

Is it Luted, or is it Bonded?

- Restoration Type/Material?
 - Emax vs PFM
 - Zirconia?
 - Implants?
- Cement Brittleness?
- Cement Texture/Color?
 - Chalky margin = brittle cement (RMGI/GIC) → low fracture toughness → easier ultrasonic removal.
 - Glossy resin margin = high-hardness adhesive cement → must section due to high cohesive strength.
- Err on the Side of Caution: Treat as Luted or Bonded?





Removal strategies:

Step 1: Determine Luted vs. Bonded

- Luted:

- Mechanical retention from prep geometry + brittle cement = cohesive failure
- More predictable removal
- Ultrasonic or mechanical crown removers

- Bonded:

- Micromechanical + chemical bond
 - Stronger retention: higher cohesive strength
 - High hardness & high fracture toughness
- Disrupt resin–tooth interface
- Section with fine diamond
- Risk: enamel or dentin fracture





You're a *faculty member*. A student asks you to verify some claims—would you say it's a MYTH or a FACT?



BO3: Explain how restorative material properties (such as brittleness, hardness, translucency, and fracture behavior) affect the selection and effectiveness of removal techniques.

Navid Khorashahi

Faculty Mentors: Dr. Britton and Dr. Harris

Fracture toughness/Brittleness:

 Low Fracture Toughness Brittle behavior — cracks initiate and propagate easily	 High Fracture Toughness Tougher material — resists crack propagation better
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HIGHEST

Zirconia

Highest fracture toughness of all dental ceramics

Clinical Significance:

Most resistant to fracture under occlusal forces. Ideal for posterior restorations and bridges.



MODERATE

Lithium Disilicate

Intermediate toughness — less than zirconia, more than feldspathic ceramics

Clinical Significance:

Good strength with excellent esthetics. Suitable for anterior and posterior crowns with adequate thickness.



LOWEST

Feldspathic Ceramics

Lowest fracture toughness compared to zirconia and lithium disilicate

Clinical Significance:

Most prone to chipping. Best for veneers and anterior crowns in low-stress situations.





Table 1: Flexural strength of pressed feldspathic ceramic, lithium disilicate ceramics, and CAD/CAM-milled zirconia bars—control group

<i>Type of ceramic</i>	<i>Groups</i>	<i>Flexural strength (MPa)</i>
Feldspathic	I	79.7
Lithium disilicate	II	221.7
CAD/CAM-milled zirconia	III	800.2



Hardness



Hardness determines how easily burs can abrade the material during crown preparation and adjustment.

Harder materials require diamond grit burs; softer materials can be cut with carbide burs.

Zirconia

HARDEST

Highest on Vickers Hardness scale

Clinical Implication:

Requires diamond burs for all adjustments. Abrades burs most rapidly.

Lithium Disilicate

MODERATE

Less hard than zirconia; cuts more readily

Clinical Implication:

Easier to adjust than zirconia but still requires diamond burs.

Metal Substructure

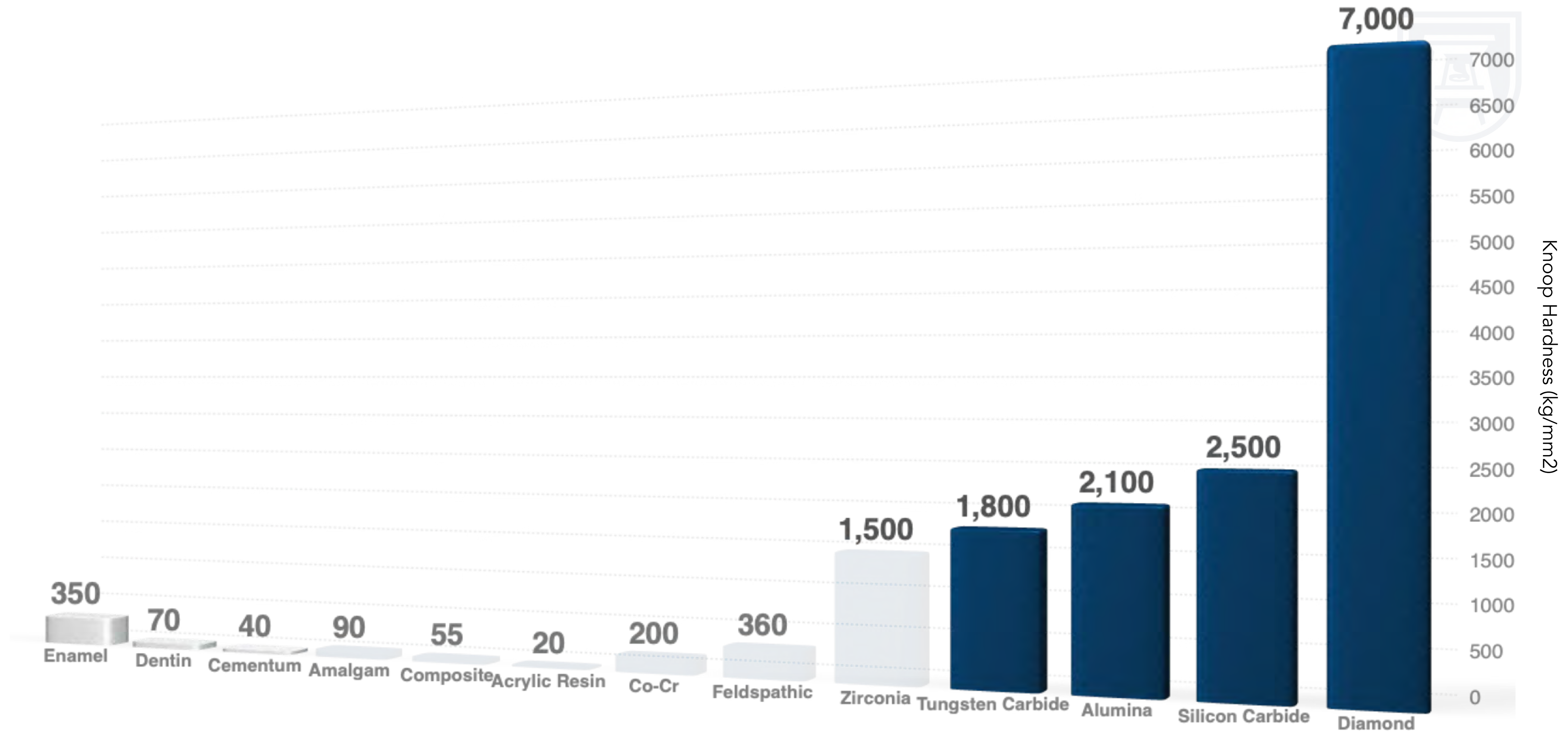
SOFTTEST

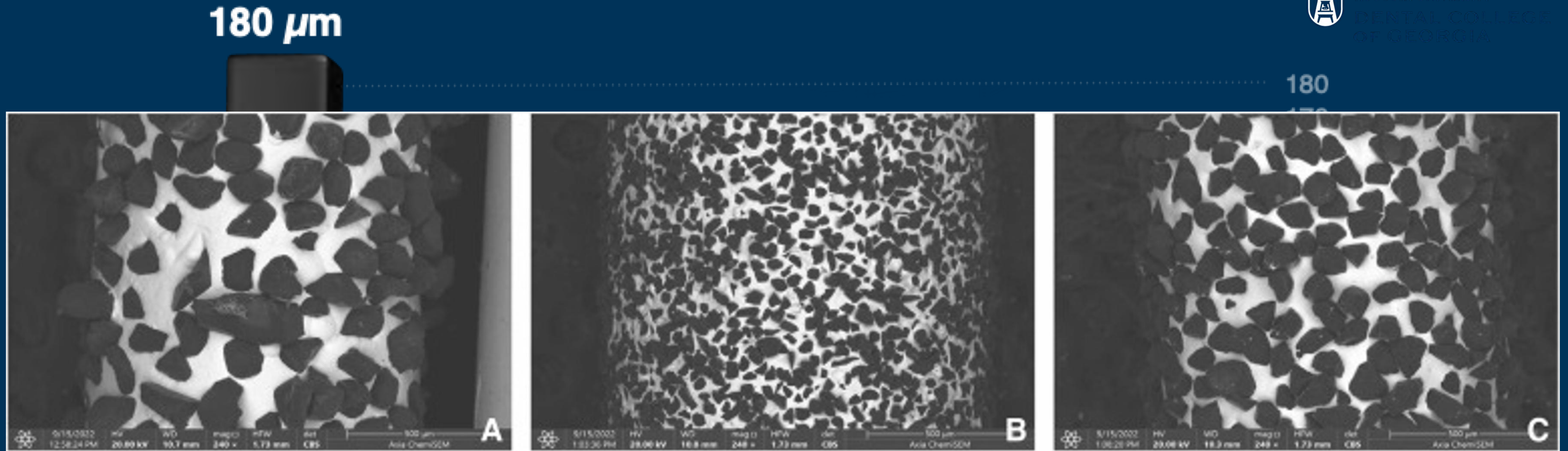
Gold and base metal alloys are softest crown materials

Clinical Implication:

Can be efficiently cut with carbide burs. Easiest to adjust.







Scanning electron micrograph showing different diamond grit of diamond rotary instruments. (Axia ChemiSEM; Thermo Fisher Scientific Inc). A, Coarse rotary instrument. B, Fine rotary instrument. C, Medium rotary instrument.

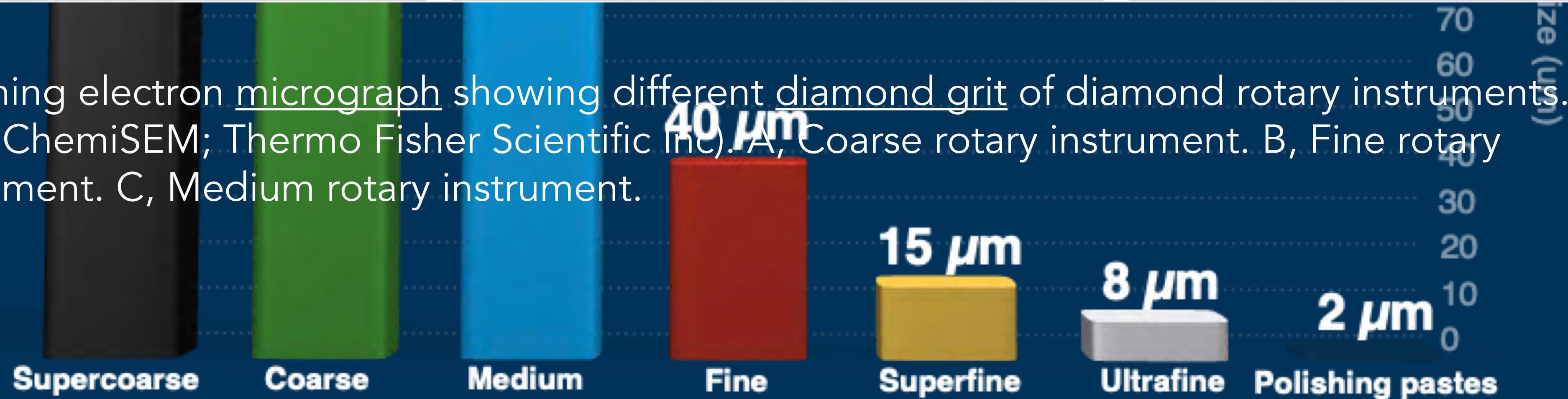




Table 6. Cutting depth (mm) of diamond rotary instruments into zirconia specimens

Instrument Grit Size Group		Cutting Depth (mm)		
Rotary Instrument	n	Mean±Standard Deviation	Median	Minimum / Maximum
Super fine (SF)	6	1.39±0.52	1.17	0.93 / 2.05
Fine (F)	6	5.79±1.44	5.87	3.64 / 7.38
Medium (M)	6	4.86±0.41	4.75	4.48 / 5.47
Coarse (C)	6	4.54±1.22	4.46	2.70 / 6.49
Super coarse (SC)	6	4.80±0.75	5.00	3.54 / 5.44



Instrument Diameter (mm) and Deterioration (%)

Rotary Instrument Grit	n	Mean±Standard Deviation	Median	P*					
									
Super fine (SF)									
First reading (mm)	6	1.33±0.00	1.33	.003 /.031	Coarse (C)				
Second reading (mm)	6	1.30±0.01	1.31		First reading (mm)	6	1.35±0.01	1.35	
Deterioration (%)	6	2.01±0.91	1.88		Second reading (mm)	6	1.25±0.03	1.25	
Fine (F)					Deterioration (%)	6	7.06±1.99	7.41	<.001 /.031
First reading (mm)	6	1.43±0.01	1.43	<.001 /.031	Super coarse (SC)				
Second reading (mm)	6	1.37±0.01	1.37		First reading (mm)	6	1.34±0.01	1.35	
Deterioration (%)	6	3.97±0.83	3.85		Second reading (mm)	6	1.22±0.03	1.22	
Medium (M)					Deterioration (%)	6	9.05±2.44	9.33	<.001 /.031
First reading (mm)	6	1.35±0.00	1.35						
Second reading (mm)	6	1.31±0.01	1.32						
Deterioration (%)	6	2.84±0.87	2.59	.001 /.031					

Special "Crown Cutting" Burs:

Extra-Coarse - Purple



Particle grit size: 230 microns



Komet 4ZR
Coarse-Grit Diamond



Translucency:



Translucency allows light to pass through the material, making the boundary between restoration and tooth/cement less visible.



OPAQUE

Zirconia

Least translucent (most opaque) material

Clinical Advantage:

Excellent visibility of margins during preparation. Easy to distinguish from tooth structure. Can mask dark substrates.



MODERATE

Feldspathic Ceramics

Moderate translucency with good light transmission

Clinical Consideration:

Balanced esthetics and visibility. Reasonable margin definition during sectioning. Natural appearance.



HIGHEST

Lithium Disilicate

Most translucent material, closest to natural tooth structure

Clinical Challenge:

Difficult to distinguish restoration from tooth during sectioning. Requires extra care to avoid over-preparation. Superior esthetics.





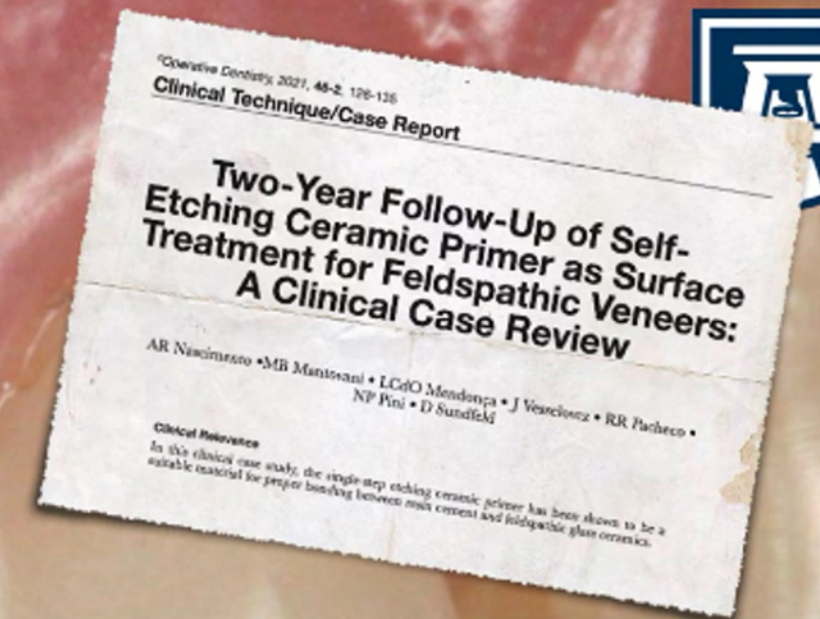
What's the issue with translucency?

Increased difficulty with distinguishing between cement, core build-up, natural tooth and crown.

What can be done to solve this?

- Using an explorer to feel and scrape away material can help.
- Aids in distinguishing cement from core/crown.
- Using a non-traumatic removal technique







Operative Dentistry, 2021, 46(2), 126-135
Clinical Technique/Case Report

Two-Year Follow-Up of Self-Etching Ceramic Primer as Surface Treatment for Feldspathic Veneers: A Clinical Case Review

AR Nascimento • MB Mantovani • LCdO Mendonça • J Vesselovcz • RR Pacheco •
NP Pini • D Sundfeld

Clinical Relevance
In this clinical case study, the single-step etching ceramic primer has been shown to be a suitable material for proper bonding between resin cement and feldspathic glass ceramic.

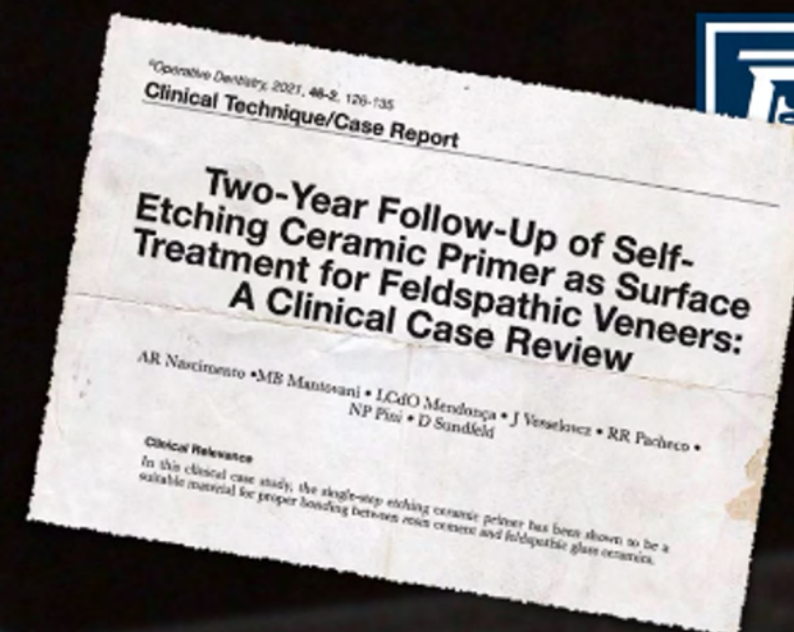


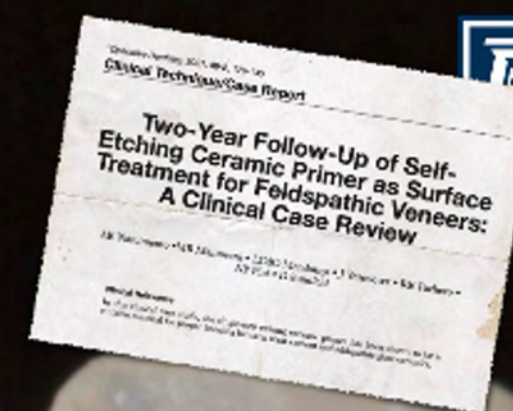
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Nascimento AR, Mantovani MB, Mendonça L, Vesselovoz J, Pacheco RR, Pini NP, Sundfeld D. Two-Year Follow-Up of Self-Etching Ceramic Primer as Surface Treatment for Feldspathic Veneers: A Clinical Case Review. Oper Dent. 2021 Mar 1;46(2):126-135. doi: 10.2341/20-036-T. PMID: 34192329.



Ceramic Crowns

HARDER
MATERIAL

- **Bur Selection**

Use fine or super-fine diamond burs with copious water irrigation

- **Technique**

Work slowly and methodically — ceramics are hard and require patience

💧 *Adequate irrigation prevents heat buildup and bur wear*

- **Special Considerations**

High translucency makes margins difficult to see

If sectioning proves too difficult, consider alternative removal methods (ultrasonic, crown removal instruments)

Metal Crowns

SOFTER MATERIAL

- **Bur Selection**

Use carbide burs with copious water irrigation

- **Technique**

Cuts more easily than ceramic — still requires controlled, careful sectioning

💧 *Irrigation prevents heat damage to pulp and surrounding tissues*

- **Advantages**

Easier to distinguish from tooth structure and cement due to opacity

Alternative removal methods (slotting, tapping off) may be more practical than complete sectioning



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Universal Principles

 **DO NOT RUSH** — Take time to identify what you're removing

 **Be patient** — Don't forcefully try to cut through resistant material



Imagine you're on a company's product design team. You're pitching a new bur/removal tool to your boss.





“Our Black Cobras are durable, coarse diamonds that we’ve found are very effective at cutting through hard materials like Zirconia.” - Meisinger USA

“Deep Purple™ Diamonds are **extra-coarse** and designed for fast, effective substance removal. Rapid and reliable substance removal due to high quality and uniformly sized extra-coarse diamonds” - Komet USA

BO4: Compare and contrast varied techniques for removing crowns, onlays, and other indirect restorations (e.g., sectioning with burs, ultrasonic vibration, crown pullers, laser-assisted methods) by explaining their mechanisms of action and contrasting their advantages and limitations based on scientific evidence.

Ethan Walden & Drew Mancao

Faculty Mentor: Dr. Britton and Dr. Harris



Search Strategy



P: Removal of insufficient indirect restoration

I: Various techniques

C: Non-removal

O: Atraumatic removal of indirect restoration.

Databases used

Pubmed

Google Gemini

Science Direct

Keywords

Indirect restoration removal strategies

Pubmed search string/Gemini Query

((Indirect) AND (Restoration)) AND (Removal) AND (Methods)

Search results

28 articles found though query

11 articles used for research

Articles sorted by information found in gemini query

Highest level of evidence

Systematic reviews, guidelines, systematic reviews of randomized clinical trials



Before we begin...



Patient Chief
Concern

Safety

Clinical
Considerations

Preserve/reuse?

- Conservative vs
Destructive

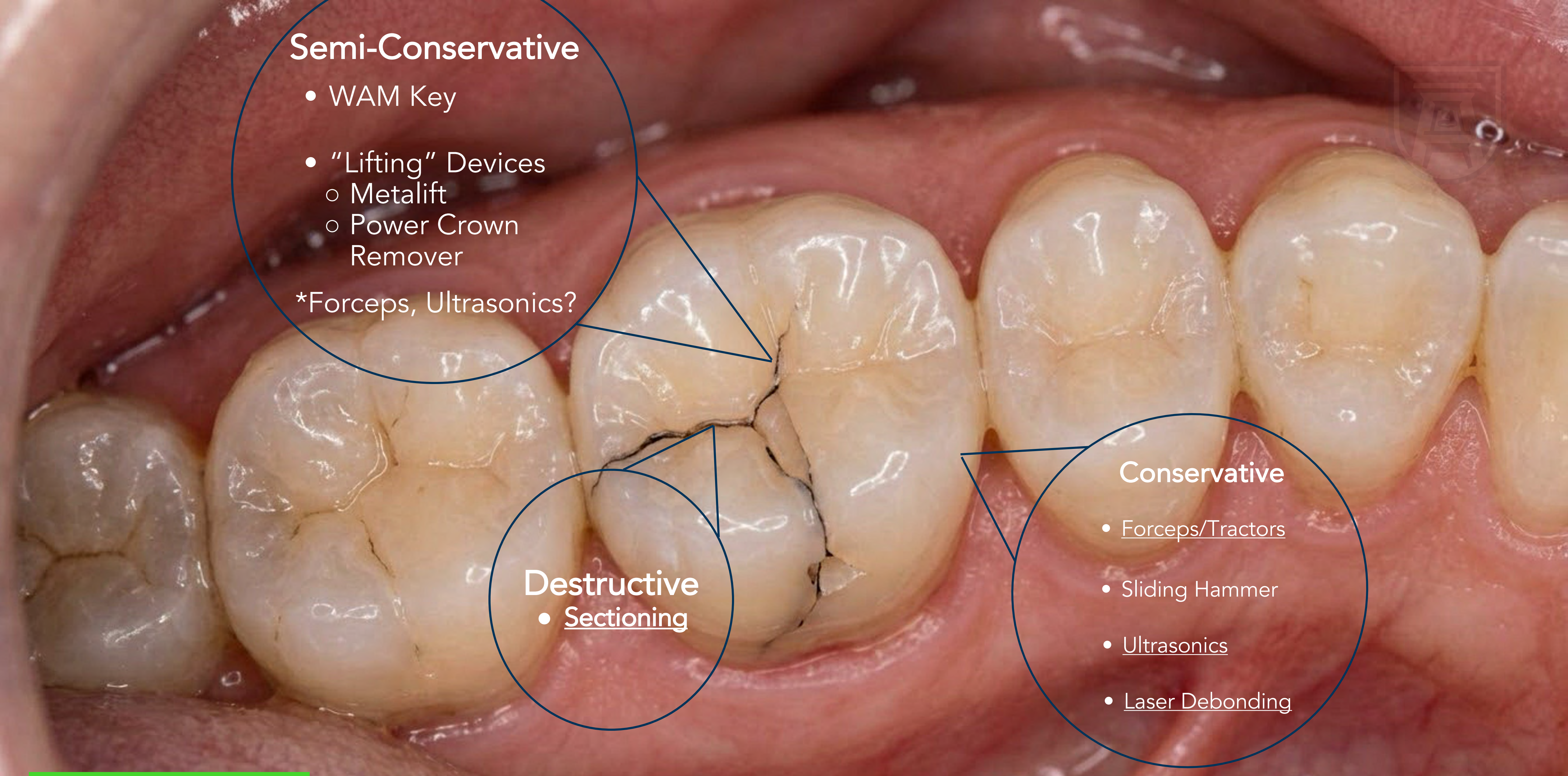
Material

- Gold, Zr, PFM...

Retention

- Luted (Conventional)
- Bonded (Adhesive)





Semi-Conservative

- WAM Key
- “Lifting” Devices
 - Metalift
 - Power Crown Remover

*Forceps, Ultrasonics?

Destructive

- Sectioning

Conservative

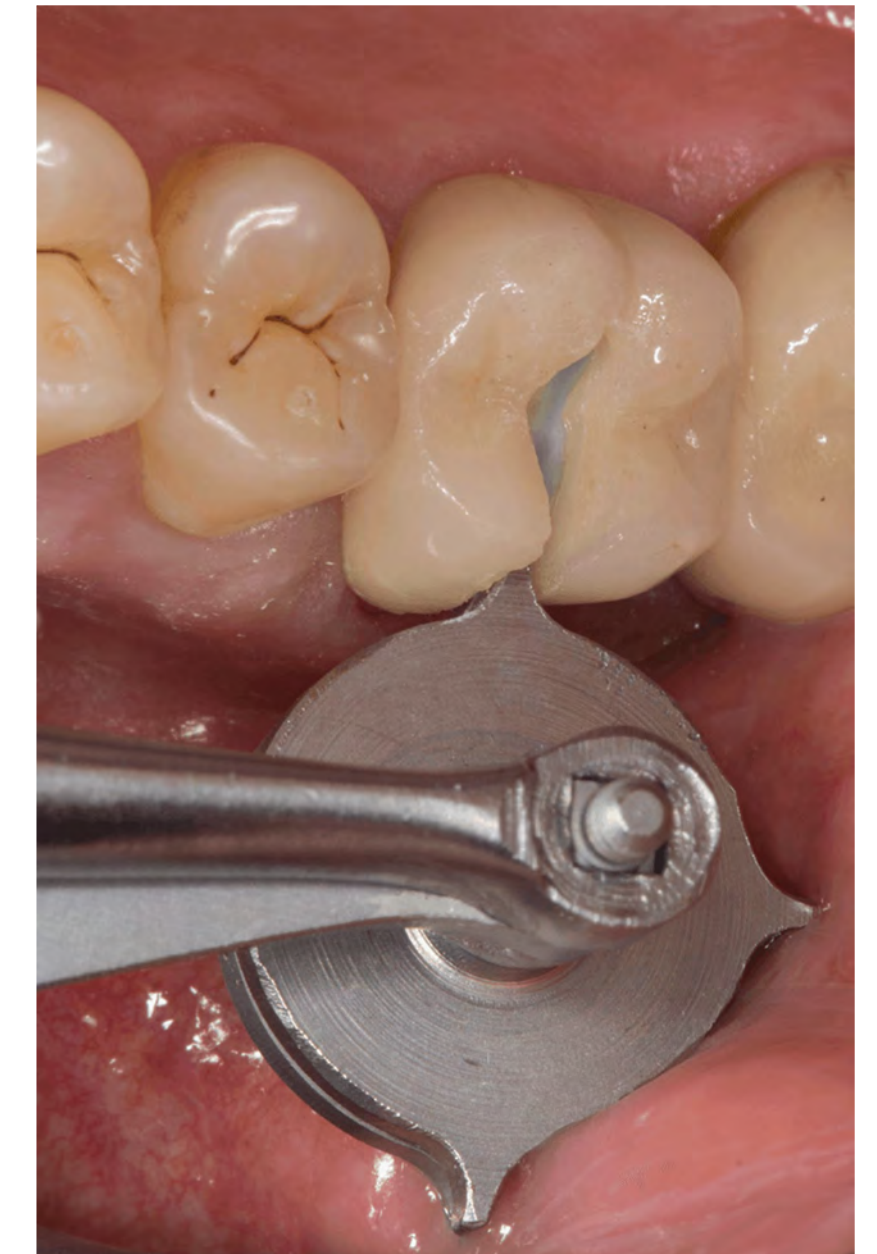
- Forceps/Tractors
- Sliding Hammer
- Ultrasonics
- Laser Debonding



Sectioning



- **Advantages:**
 - Fastest method, lowest technical sensitivity, "tried and true"
- **Disadvantages:**
 - Heat, iatrogenic trauma are the largest concern → Use metal partitioning to protect adjacent teeth
 - "Long-term biologic trade off" → Potential to damage abutment tooth
- **Indications:**
 - Monolithic Zr crowns that **are not** being reused, Stubborn PFM/Metal crowns (last resort)





- **MOA:**
 - Placement of 1-2 "K-grooves" (a stress concentration point in the restoration), usually on the B and L of the crown
 - Followed by usage of crown spreader to break cement



Can get expensive with increased use
of burs

"Expect to go through 5-7 burs per zirconia crown" -
Dr. Pacheco

- Burs

- FG Carbide Crown Cutting burs:
 - (1931) - pack of 5 \$46.30 on the dental supply company .com
- Zirconia Cutting
 - Great white zirconia kit - \$71.95 for a kit of 6 burs for zirconia removal - Dental wholesale direct .com



Ultrasonic



- **Advantages**
 - Atraumatic, speed
- **Disadvantages**
 - Heat generation→ Potential to cause pulpal or structural trauma
 - Can be technique sensitive (use light touch, constant movement)
- Best for: Metal/PFM crowns, LiSi veneers when reuse is desired

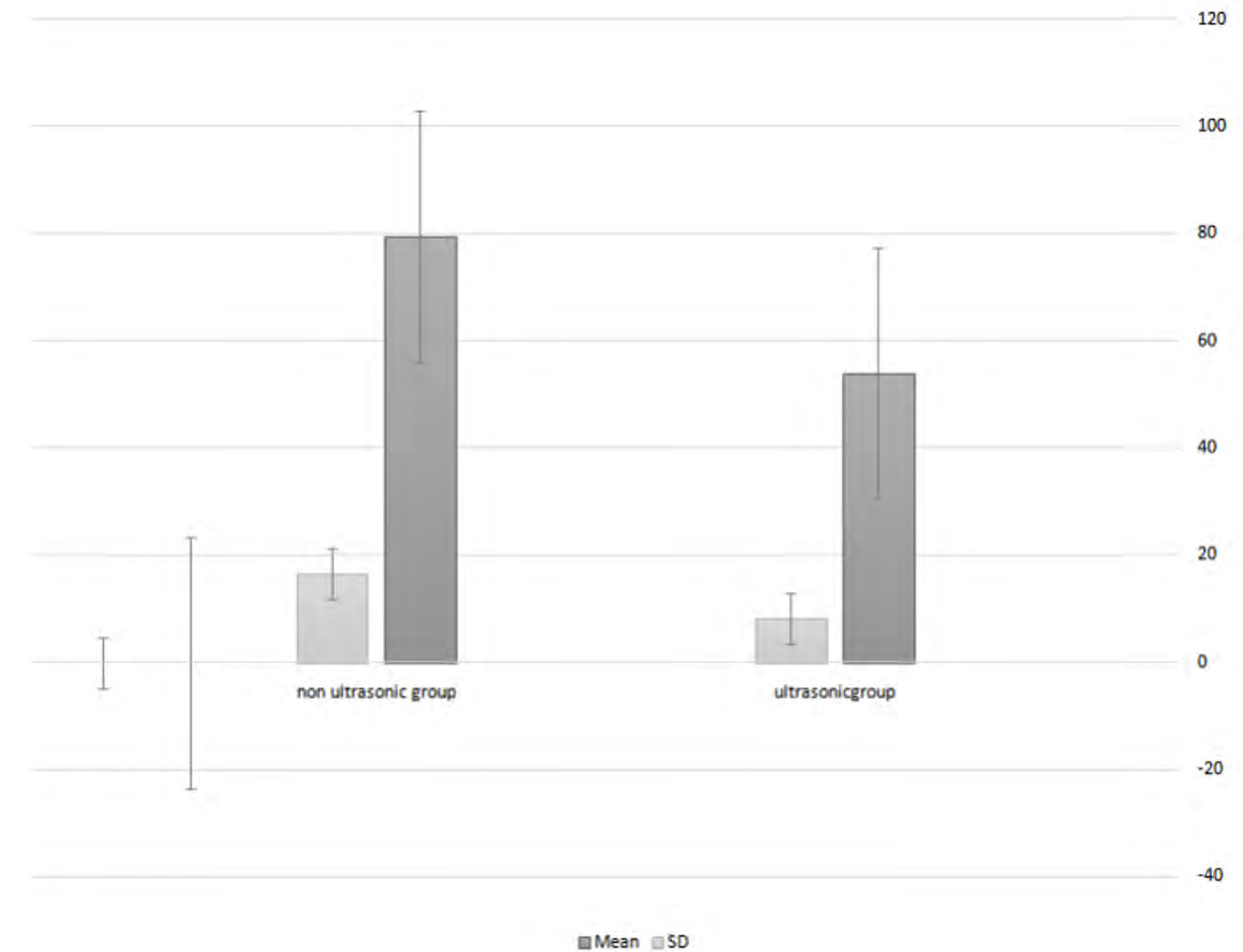


- MOA

- Electrical → mechanical energy
- High frequency vibration causes fatigue and mechanical fractures in the cement which can lead to the separation of the indirect restoration



- Units range from low end (\$150-300) to high end (\$1k-3k)
- Tip cost ~\$30



Mechanical Pullers



- MOA
 - Deliver controlled tensile or impact force to weaken and break mechanical retention of conventional cements or luting agents.
- Best for:
 - Old PFM and Metal crowns that utilize Zinc phosphate or Polycarboxylate for retention.
 - Temporary crowns





WAMKEY BRIDGE & CROWN REMOVER

Mechanical Pullers



- **Advantage**
 - Simple, effective, low-cost, fast
- **Disadvantage**
 - Potentially traumatic to tooth structure(s): root, periodontium, adjacent crowns/abutments
 - Potentially Stressful for patients. Lots of pressure, mallet type- impact is jarring
- **Cost:** Relatively inexpensive. Kits range from \$50-\$200 dollars

Laser Debonded

- MOA

- Er:YAG laser set to particular wavelength (2940nm) and power setting
- Semi-translucent and translucent crown materials such as Zr, LiDi do not absorb much* of the 2940 nm wavelength beam
- Laser is absorbed by the water and organic components of the cement or luting agent, weakening it and making it susceptible to gentle mechanical removal with crown pullers (thermal softening and selective ablation)



- Best for:
 - **Complex Zirconia** (natural teeth/ implant abutment) and Lithium Disilicate crowns



Laser Assisted

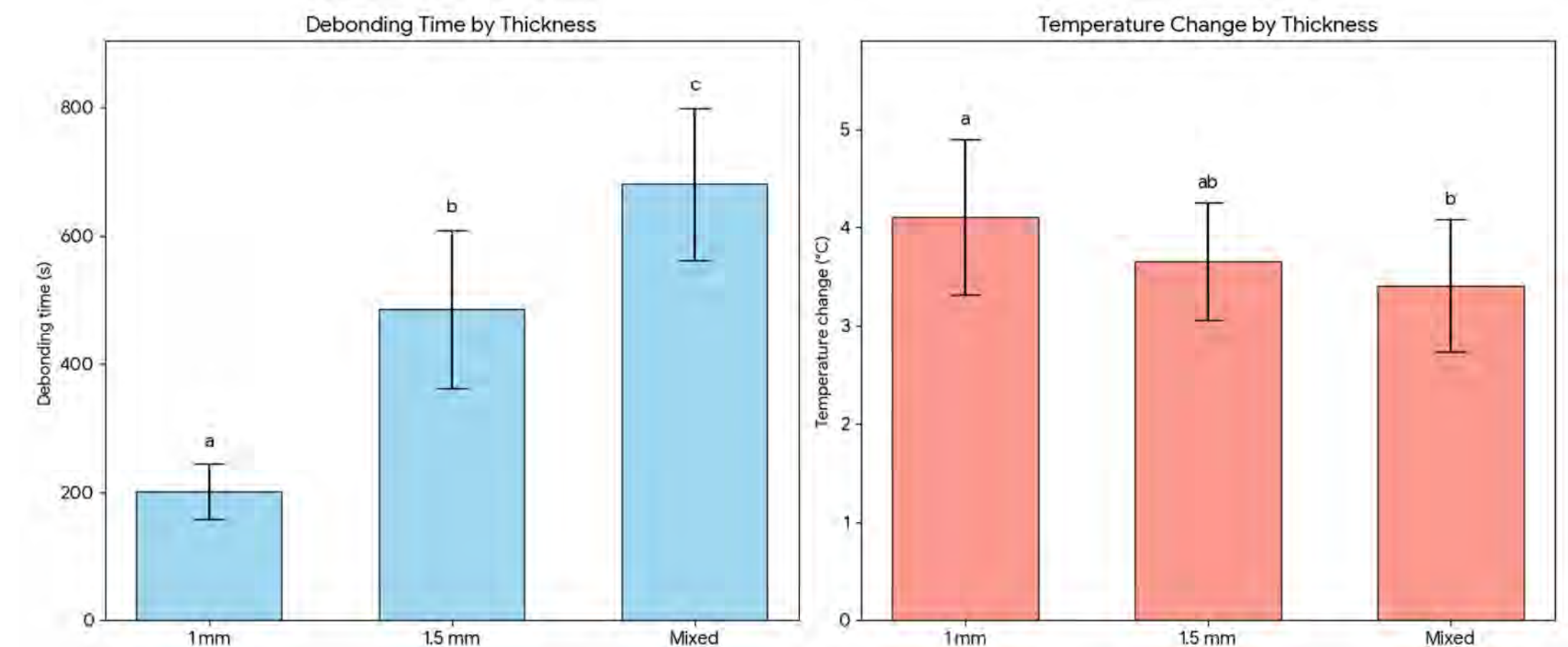
- **Advantage**

- Atraumatic removal of crowns requiring otherwise destructive methods
- Relatively fast (5-10 min)
- Low thermal rise (limited to 3-5 C)
- 95% success rate clinical/in-vitro study

- **Disadvantage**

- Probably sticking a laser in someones mouth
- Thermal rise (if poorly controlled)
- Highly technique sensitive.
- COST

- **Cost:** **MOST EXPENSIVE**: Ranging from \$12k-\$40k for lasers graded for this purpose, handpieces for the laser





Zirconia crown removal with LibeTouch™ Er:YAG laser
Dr. Sasha Donyanov & Dr. Tony Carreon





Technique	Mechanism of Action (MOA)	Advantages	Disadvantages
Sectioning with Burs	Bur placement on Buccal & Lingual followed by crown spreader to break cement.	Fastest method Least technique sensitive	Potential for heat generation Iatrogenic trauma to abutment
Ultrasonic Vibration	Rapid vibration at margins causes fatigue and failure of cement.	Atraumatic Rapid speed	Heat generation Technique sensitive
Mechanical Pullers	Controlled tensile force breaks mechanical retention of cement.	Low cost Fast Simple procedure	Traumatic to tooth structure Patient discomfort (pressure)
Laser Debonding	Er:YAG laser (2940nm) absorbs into water/organic components, weakening cement.	Relatively atraumatic Fast (5-6 min) Low thermal rise (if controlled)	Highly technique sensitive Requires strong laser in mouth Thermal rise (if uncontrolled)



Insurance Approval Challenge

A patient's **zirconia crown on #14** is fractured and needs replacement.

The dentist must remove what's left of the old crown.



BO5: Apply a biologically informed clinical decision tree to select the most appropriate removal method, integrating material type, retention strategy, and mechanical/optical properties while prioritizing tooth preservation and patient comfort.

Austin Nguyen & Anthony Turner II

Faculty Mentor: Dr. Britton and Dr. Harris



Search Strategy



PICO: Not Applicable

Databases used

Pubmed

Keywords

Luted, Bonded, Crown, Removal,
Methods, Laser, Ultrasonic,
Instruments

Pubmed search string

((Remove) AND (Luted Crown)) AND
((Remove) AND (Bonded))

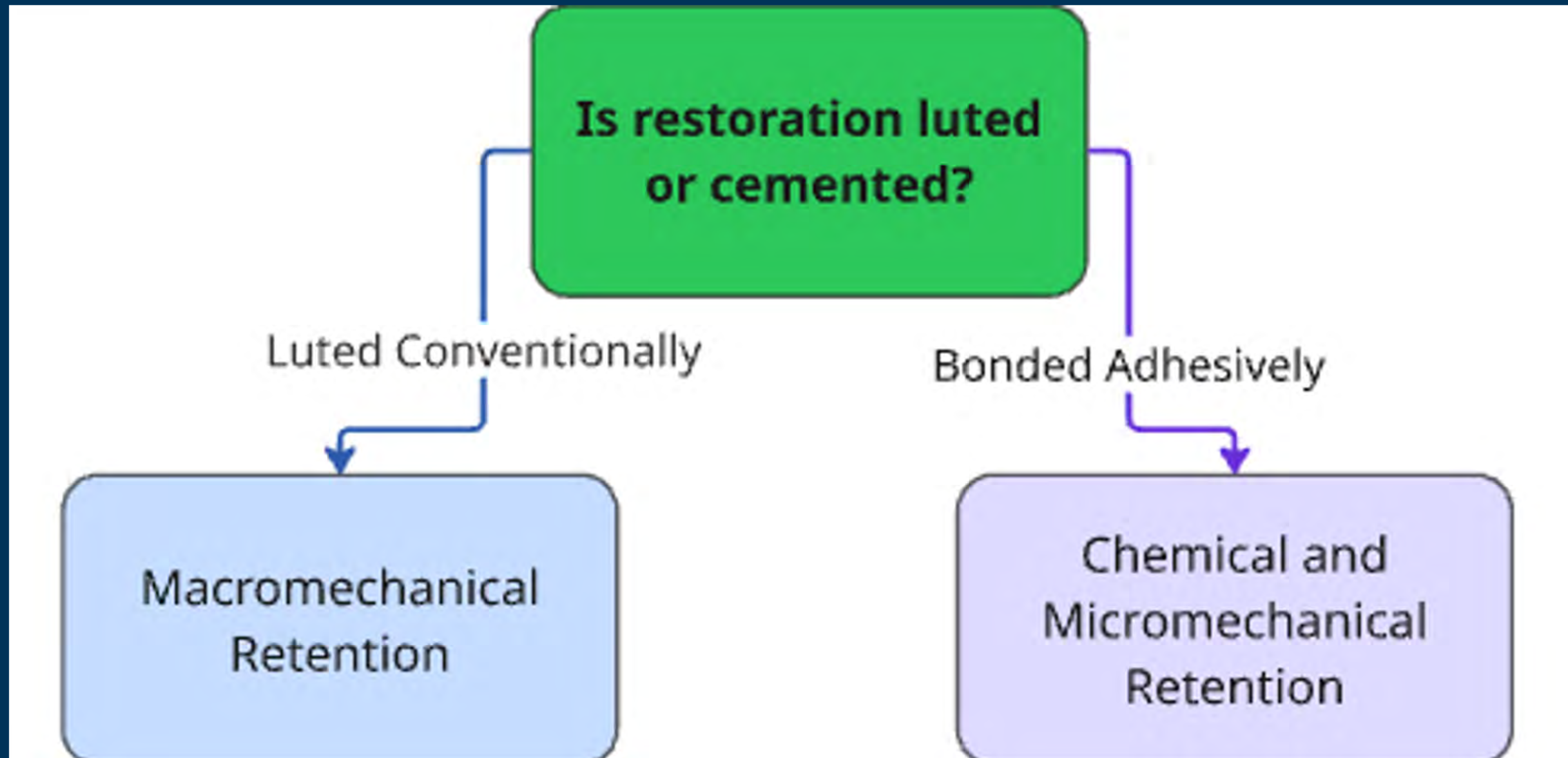
Search results

240 articles found though query

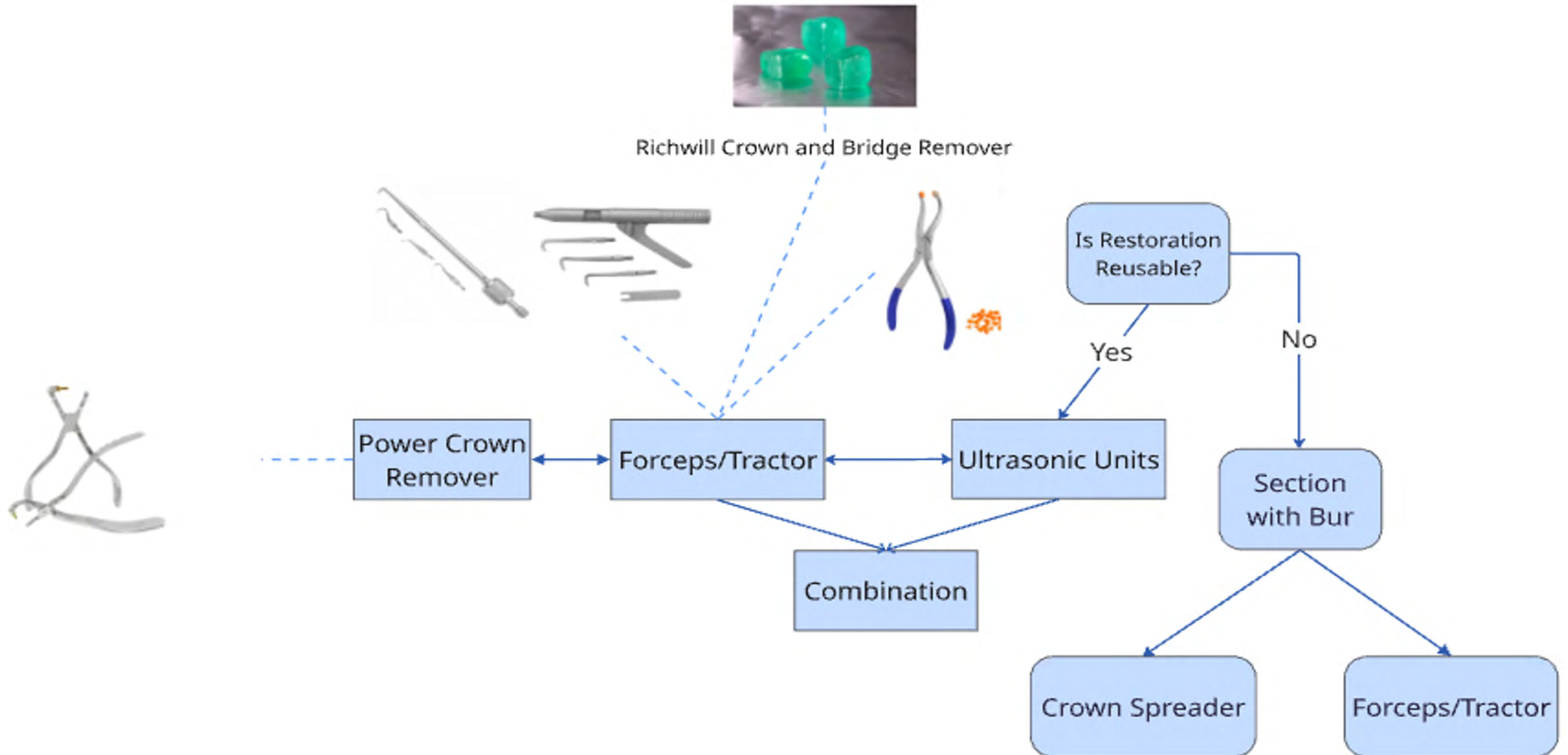
Highest level of evidence

Systematic reviews

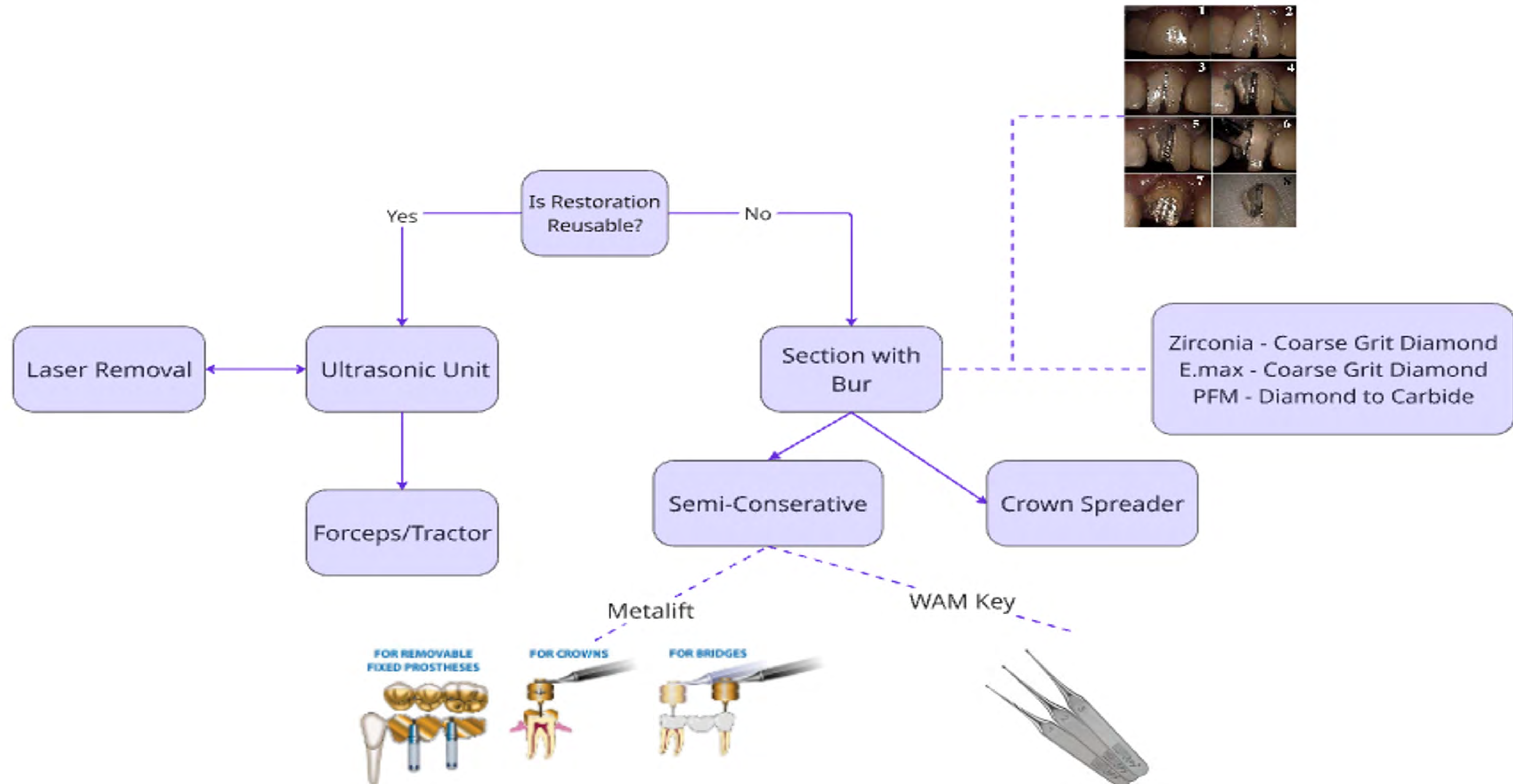
Removal Process



Luted Conventionally



Bonded Adhesively





Worst-Case Scenario Challenge

You are a *dentist* (you in 1 ½ years)!

You are removing a zirconia crown on #30 with a Komet Deep Purple bur under high water irrigation.





Dr. O'Donnell "It takes 5 minutes to cut it off with diamond flame and spread it with an elevator or screwdriver to remove."

Dr. Crumpton "In my mind, I don't think speed - I think efficiency." Let the bur do the work (reducing trauma and excess reduction).



Six Thinking Hats®



"Feel like a patient."



"Judge like the board."



"Think like a faculty member."



"Innovate like a manufacturer."



"Worry like insurance."



"Work like a dentist."

Behavioral Objective 1: Describe the biological and mechanical risks of removing indirect restorations, including potential damage to enamel, dentin, pulp, and supporting tissues.

- The removal of indirect restorations is a routine dental procedure that can be challenging as there are mechanical(fractures, iatrogenic gouging) and biological risks(iatrogenic heat, soft tissue trauma) to be aware of.
- Dentists must choose the appropriate removal method based on the restoration type and tooth status as well as prioritizing THERMAL CONTROL and STRUCTURAL PRESERVATION to ensure long-term success as removing the old crown is only your first step in fabricating a new crown for your patient.
- Remember to always use copious amounts of water to for cooling, HVE suction, adequate light and visualization, and always make the patient aware of what is going on as this can be a loud and uncomfortable procedure.





Behavioral Objective 2:

Differentiate between restorations that are bonded adhesively versus those luted conventionally, and explain how the mode of retention influences removal strategies.

- Conventional luting uses mechanical retention and brittle cements, such as RMGI or GI. These cements allow restorations to be removed predictably with ultrasonic or mechanical tools that break the cement layer. On the other hand, adhesively bonded restorations use resin cements that create chemical and micromechanical bonds with both the tooth and the restoration. This provides stronger retention but often requires the restoration to be sectioned for removal to prevent damage to the tooth. Understanding these differences can help providers choose the least invasive and traumatic way to remove crowns, onlays, or other indirect restorations.





Behavioral Objective 3: Explain how restorative material properties (such as brittleness, hardness, translucency, and fracture behavior) affect the selection and effectiveness of removal techniques.

- For ceramic crowns, it's best to use fine or super-fine diamond burs with ample irrigation, avoid forcing the bur, and take your time to clearly identify what you're removing; if sectioning becomes too difficult, alternative removal methods may be appropriate. For metal crowns or metal substrates, carbide burs with generous irrigation are preferred, as the material is easier to distinguish from tooth structure and cement, and other removal methods may also be suitable when needed.





Behavioral Objective 4: Compare and contrast varied techniques for removing crowns, onlays, and other indirect restorations by explaining their mechanisms of action and contrasting their advantages and limitations based on scientific evidence.

- Sectioning with burs is fast and simple but carries risks of heat and iatrogenic damage. Ultrasonic vibration offers an efficient, generally atraumatic option but is technique-sensitive and can generate heat. Mechanical pullers are low-cost and quick but can be traumatic and uncomfortable for patients. Laser debonding is relatively atraumatic and fast with controlled thermal rise, but it is highly technique-sensitive and involves applying a strong laser in the mouth.



Behavioral Objective 5: Apply a biologically informed clinical decision tree to select the most appropriate removal method, integrating material type, retention strategy, and mechanical/optical properties while prioritizing tooth preservation and patient comfort.

- Following the flowchart that was created, it is important to determine whether a restoration is luted or bonded when determining way of removal. There are different strategies that can be used each with their pros and cons. In terms of luted restorations, the most conservative approaches look to preserve the restoration and mitigate damage to tooth structure. When a restoration cannot be reused and the approach with ultrasonic and mechanical instruments fails, a technique with sectioning is used. On the other hand, bonded restorations mostly require sectioning due to the chemical and micromechanical retention that is used. A way to preserve the restoration would be using a laser to break the bonds holding the restoration and tooth together. What was found on the DCG protocol for bonding lithium disilicate and zirconia was the research that we found to be the best for retention in terms of cost, effectiveness, and being user-friendly.



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